OPTIMAL TEMPERATURE PROFILE FORMATION FOR BIODIESEL PRODUCTION IN A BATCH REACTOR WITH UNCERTAINTY – STOCHASTIC CONTROL

A Project Report submitted in partial fulfilment
of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
In
Electronics and Instrumentation Engineering

Submitted by

Tanish J

Reg. No.: 190932190

Under the guidance of

Dr. I. Thirunavukkarasu
Professor
Instrumentation and Control Engineering
Manipal Institute of Technology

DEPARTMENT OF INSTRUMENTATION AND CONTROL ENGINEERING

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SEPTEMBER 2023

DEPARTMENT OF INSTRUMENTATION AND CONTROL ENGINEERING

MANIPAL INSTITUTE OF TECHNOLOGY

(A Constituent Institution of Manipal Academy of Higher Education)

MANIPAL – 576 104 (KARNATAKA), INDIA

Manipal
05-Sep-2023

CERTIFICATE

This is to certify that the project titled **OPTIMAL TEMPERATURE PROFILE FORMATION FOR BIODIESEL PRODUCTION IN A BATCH REACTOR WITH UNCERTAINTY – STOCHASTIC CONTROL** is a record of the bonafide work done by **TANISH J** (*Reg. No. 190932190*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (BTech) in **ELECTRONICS AND INSTRUMENTATION ENGINEERING** of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent Institution of Manipal Academy of Higher Education), during the academic year 2022-2023.

Dr. I. Thirunavukkarasu
Professor
Dept. of Instrumentation
and Control Engineering
MIT Manipal

Prof. Dr. Shreesha C
Head of the Department
Dept. of Instrumentation
and Control Engineering
MIT Manipal

ACKNOWLEDGMENT

First and foremost, I would like to thank my project guide, Dr. I. Thirunavukkarasu Sir, for giving me the opportunity to take up this project. His exceptional expertise and mentorship were pivotal in the success of my project. Your belief in me propelled me to excel. I greatly appreciate the contributions of my HOD, Dr. Shreesha Sir who always gave me valuable inputs during the project. I am also very thankful to my family and friends for their unwavering support and motivation.

This project has been an enriching experience, greatly expanding my understanding in the field of control systems and optimal control.

ABSTRACT

Biodiesel production is a promising renewable energy source that can reduce greenhouse gas emissions and dependence on fossil fuels. However, inherent uncertainties in the process can affect product quality, quantity, and process economics. Two of the most common types of uncertainty in biodiesel production are variability in feed composition and reactor operation. To address these uncertainties, mathematical models have been developed that simulate the transesterification process and account for the variability in feed composition and reactor operation.

Stochastic differential equations (SDEs) or Ito processes are used to model the uncertainty in the feed composition. SDEs account for the variability in the feed composition and simulate the expected concentration of methyl ester in the biodiesel product. Ito's Lemma is used to calculate the expected concentration of methyl ester in the biodiesel product given the uncertainty in the feed composition. Stochastic Maximum Principle is used to find the optimal temperature profile for the batch reactor, given the uncertainty in the feed composition and reactor operation. The optimal temperature profile maximizes the expected concentration of methyl ester (E) in the biodiesel product, while satisfying the constraints of the reactor operation. The optimal temperature profile can be obtained iteratively by solving the Hamiltonian Equation using numerical methods, such as the finite difference method or the finite element method.

Closed-loop operation involves monitoring the reactor operation and adjusting the temperature profile in real-time to ensure that the system operates optimally. The closed-loop operation can be implemented using a feedback control system, which adjusts the temperature profile based on the current state of the reactor. By addressing the uncertainties in feed composition and reactor operation, the use of these mathematical models and control strategies can improve the yield and quality of biodiesel production, making it a more sustainable and efficient energy source.

In this project, the uncertainties in feedstock composition have been accounted for using Ito's Lemma and Ito's Calculus on MS Excel. The plots of concentration of the 6 elements of biodiesel have been plotted using experimental data before and after the application of Ito's process. Stochastic Maximisation Principle has been applied onto the Objective Function and uncertainties added to obtain the adjoint equations for the elements of biodiesel and their respective uncertainties. The adjoint equations have been solved using MATLAB programs and Hamiltonian is solved and optimal plot is obtained.

TABLE OF CONTENTS

			Page No
Acknowledgement			iii
Abstract			iv
List of Figures			vi
Chapter 1		**INTRODUCTION**	
	1.	Introduction	1
	2.	Rationale	3
	3.	Objectives	4
	4.	Apparatus	4
Chapter 2		**LITERATURE REVIEW**	
	1.	Present state in area of work	5
	2.	Background Theory	5
	3.	Literature Survey	6
	4.	Literature review summary	7
Chapter 3		**METHODOLOGY**	
	1.	Batch Reactor Model for Biodiesel	8
	2.	Ito's Process	9
	3.	Optimal Control Problem	16
	a)	Stochastic Maximum Principle	16
	b)	Ito's Lemma	17
	c)	The Hamiltonian	18
Chapter 4		**SIMULATION**	
	1.	MATLAB Simulations	27
Chapter 5		**RESULTS**	
	1.	Deterministic vs Stochastic Cases	33
	2.	Results Analysis	34
Chapter 6		**CONCLUSION**	36
REFERENCES			37

LIST OF FIGURES

Figure No.	Figure Title	Page No
1	Image of a Lab Scale Batch Reactor	4
2.1	Concentration of Triglycerides before applying Ito's Process	10
2.2	Concentration of Triglycerides after applying Ito's Process	10
2.3	Concentration of Diglycerides before applying Ito's Process	11
2.4	Concentration of Diglycerides after applying Ito's Process	11
2.5	Concentration of Monoglycerides before applying Ito's Process	12
2.6	Concentration of Monoglycerides after applying Ito's Process	12
2.7	Concentration of Methyl Ester before applying Ito's Process	13
2.8	Concentration of Methyl Ester after applying Ito's Process	13
2.9	Concentration of Methanol before applying Ito's Process	14
2.10	Concentration of Methanol after applying Ito's Process	14
2.11	Concentration of Glycerol before applying Ito's Process	15
2.12	Concentration of Glycerol after applying Ito's Process	15
4.1	Concentration profile at constant temperature for 100 min	27
4.2	Adjoint variable(z_i) vs. Time Plot for 100 min	28
4.3	Adjoint variable(ω_i) vs. Time Plot for 100 min	28
4.4	Theta(θ) vs Time plot for 100 min	29
4.5	Phi(Φ) vs Time plot for 100 min	30
4.6	Psi(Ψ) vs Time plot for 100 min	30
4.7	Iterative dH/dT vs Time plot for 100 min	31
4.8	Iterative Reaction temperature (Tr) vs Time plot for 100 min	31
4.9	Optimal temperature profile for maximum biodiesel production (Stochastic Control)	32
5.1	Optimal temperature profile for maximum biodiesel production (Deterministic Control)	33
5.2	Comparison of Deterministic vs Stochastic Control	34

CHAPTER 1: INTRODUCTION

1. INTRODUCTION

Biodiesel production is an intricate process encompassing a sequence of chemical reactions that convert vegetable oils, like soybean oil, into methyl esters. Biodiesel, being a renewable and sustainable energy source, offers the potential to diminish greenhouse gas emissions and decrease reliance on fossil fuels. Nevertheless, the biodiesel production process is subject to inherent uncertainties, which can impact the quality, quantity, and economic aspects of the final product.

Among the most prevalent sources of uncertainty in biodiesel production is the fluctuation in feedstock composition. The composition of triglycerides in biodiesel can vary significantly, necessitating the sampling and analysis of the probability distribution of these triglycerides. Feedstock composition exhibits substantial variability, and even within the same feedstock, the range of a single component can be extensive. Typically, this uncertainty is represented as a variance parameter, reflecting the uncertain nature of feed composition.

Another source of uncertainty in biodiesel production pertains to the operation of the transesterification reactor. Transesterification, a chemical reaction converting triglycerides into methyl esters, poses operational challenges such as temperature overshooting, internal pressure oscillations, variations in reactor conversion, and fluctuations in cooling jacket temperatures. These uncertainties can directly influence the yield and quality of the biodiesel product.

To address these uncertainties, researchers have devised mathematical models capable of simulating the transesterification process while accounting for the variability in feed composition and reactor operation. These models are typically grounded in the principles of mass and energy balances within the reactor, coupled with the kinetics governing the transesterification reaction.

One approach to modelling feed composition uncertainty employs stochastic differential equations (SDEs) or Ito processes. SDEs are mathematical expressions that elucidate how systems evolve under uncertainty, particularly relevant to feed composition within biodiesel production. Ito processes, a subset of SDEs, are extensively utilized across disciplines like finance, physics, and engineering to model random processes. Leveraging Ito's processes offers the advantage of accommodating feed composition variability and simulating the anticipated concentration of methyl ester in the final biodiesel product.

Ito's Lemma, a potent tool in stochastic calculus, proves invaluable for deriving the expected value of a function subject to a stochastic process's fluctuations. In the context of biodiesel

production, Ito's Lemma facilitates the calculation of the expected methyl ester concentration in the biodiesel product, considering the uncertainty in feed composition.

The Stochastic Maximum Principle (SMP), another formidable tool in stochastic calculus, provides a means to determine the optimal control strategy for systems operating under uncertainty. In biodiesel production, SMP can be deployed to ascertain the optimal temperature profile for the transesterification reactor, accounting for feed composition and reactor operation uncertainties. This optimal temperature profile maximizes the expected methyl ester concentration in the biodiesel product while adhering to reactor operation constraints.

The iterative derivation of the optimal temperature profile involves solving the Hamiltonian Equation, which outlines system evolution under uncertainty. The Hamiltonian Equation, a partial differential equation, can be solved via numerical techniques like the finite difference method or finite element method. The solution to the Hamiltonian Equation yields the optimal temperature profile for the transesterification reactor, serving as a reference trajectory for closed-loop biodiesel production operations. Closed-loop operation entails continuous monitoring of reactor performance and real-time adjustments to the temperature profile to ensure optimal system functioning.

2. RATIONALE

Biodiesel is an alternative, renewable and biodegradable source of energy derived from plants or animals. Using biodiesel as a fuel decreases greenhouse gas emissions by up to 86%, lowers particulate matter in the atmosphere by 47% and reduces hydrocarbon emissions by up to 67%. Due to its eco-friendly properties, production of biodiesel is very important to tackle global warming in today's world.

The project addresses the need for more efficient and sustainable production of biodiesel, which is a renewable energy source that can reduce greenhouse gas emissions and dependence on fossil fuels. Secondly, the project demonstrates the application of stochastic control theory to address the inherent uncertainties in biodiesel production, including the variability in feed composition and reactor operation.

The project proposes a novel approach for optimal control of biodiesel production, based on the Stochastic Maximum Principle (SMP) and Ito calculus. The approach can help improve the yield and quality of the biodiesel product, while reducing production costs and energy consumption. Finally, the research can contribute to the development of new models and methods for optimal control of other chemical processes, particularly those that involve uncertainty and stochasticity.

Optimal control has come into demand in the industrial and academic field as it provides useful information for designing and controlling the reaction process. usually, a solution to these problems involves finding the time dependent profiles of the control variable to optimize a particular performance index, in this case temperature profile. The final optimal profile is used for the closed loop operation of pilot plant batch reactor. Fig.1. shows this experimental setup of a lab scale batch reactor.

3. OBJECTIVES

1. Develop a mathematical model that can simulate and account for the variability in feed composition in transesterification process. Use stochastic differential equations (SDEs) or Ito processes to model the uncertainty in the feed composition and calculate the expected concentration of methyl ester in the biodiesel product.

2. Apply Ito's Lemma to calculate the expected concentration of methyl ester in the biodiesel product, given the uncertainty in the feed composition.

3. Use Stochastic Maximum Principle to find the optimal temperature profile for the batch reactor. Solve the Hamiltonian Equation which can be used as a reference trajectory in closed-loop operation for optimal biodiesel production.

4. APPARATUS

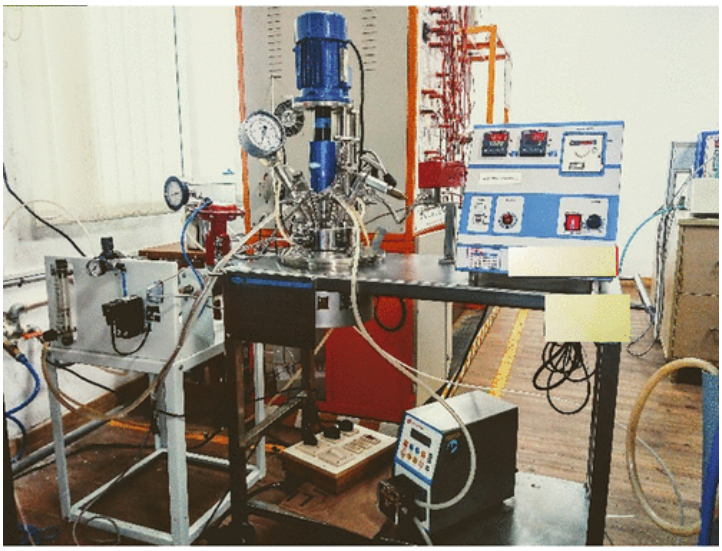

Fig. 1 Image of a Lab Scale Batch Reactor in ICE Dept, MIT Manipal

CHAPTER 2: LITERATURE REVIEW

1. PRESENT STATE IN AREA OF WORK

Despite its numerous advantages, biodiesel does come with certain technical, economic, and environmental drawbacks. In comparison to traditional fossil fuel diesel, biodiesel exhibits lower oxidative stability and cold flow properties, higher viscosity and density and increased NOx emissions. However, one of the most pressing challenges in biodiesel production is its relatively high production cost when contrasted with fossil diesel. This cost disparity primarily arises from the expensive nature of feedstocks and the inefficiencies and expenses associated with the production processes.

The prevailing and economically viable method for large-scale biodiesel production involves alkali-catalysed transesterification of fresh vegetable and animal fatty oils. Nevertheless, more than 80 percent of the production cost of biodiesel is attributed to the utilization of fresh, edible oils as feedstocks. Consequently, the most effective approach to achieving cost-effective biodiesel production entails the utilization of alternative, more affordable sources of lipids. These sources may encompass waste cooking oil, algae, municipal sewage sludge, and recycled grease trap waste.

2. BACKGROUND THEORY

The background theory of this paper is focused on the use of optimal control strategies in batch biodiesel production. The production of biodiesel involves a series of chemical reactions that transform vegetable oils into methyl esters. The paper highlights the inherent uncertainties in biodiesel production, which can impact product quality, quantity, and process economics. The authors propose an optimal control strategy that can account for the variability in feed composition and reactor operation and can help to maximise the yield and quality of the biodiesel product. The theory is based on stochastic differential equations and the stochastic maximum principle, which are powerful tools in stochastic calculus that can be used to model random processes and find the optimal control strategy for a system under uncertainty. The optimal temperature profile obtained from the solution of the Hamiltonian Equation can be used as a reference trajectory in closed-loop operation for biodiesel production. Overall, the background theory provides a foundation for understanding the importance of optimal control strategies in improving the efficiency and sustainability of batch biodiesel production.

3. LITERATURE SURVEY

Urmila Diwekar *et al* [1] describes the profile formation using Hamiltonian Equation for the biodiesel production using soyabean oil as feed. The iteration procedure for the final temperature profile is given briefly with the mathematical concept. They further considered the uncertainty in the feed component, Triglyceride. The forward and backward integration is also used to form the final temperature profile.

Urmila Diwekar *et al* [2] The paper extends the previous work of the authors on the deterministic optimal control of biodiesel production in a batch reactor. The authors have identified several challenges in biodiesel production, including the variability in feed composition, operation of the transesterification reactor, and the impact of uncertainties on process economics. The authors have also reviewed various modelling approaches, including mass and energy balances, kinetics of the transesterification reaction, and stochastic differential equations. The paper discusses the use of Ito processes and Ito's Lemma in modelling the uncertainty in the feed composition. The authors have also explained the Stochastic Maximum Principle and its application in finding the optimal temperature profile for the transesterification reactor under uncertainty. The research paper presents the optimal control strategy for biodiesel production in a batch reactor using the closed-loop control approach. The authors have implemented the feedback control system to adjust the temperature profile in real-time to ensure that the system operates optimally. The results show that the proposed optimal control strategy significantly improves the yield and quality of the biodiesel product. The authors have presented a comprehensive study of the modelling and control of the process under uncertainty. The research has practical implications for the biodiesel industry, as it can help to improve the product quality, yield, and process economics.

In both the above papers only the temperature profile formation is discussed, the control algorithms for maintaining the trajectory tracking with respect to the reactor temperature is not presented.

Prajwal Shettigar J *et al* [3] discusses data-driven modelling of a batch reactor and validation of a nonlinear model predictive controller for dynamic temperature profile tracking. In this closed loop study, the reactor temperature control has been studied by keeping the power supply to the heater as constant and manipulating the coolant flow rate. It was observed that the temperature profile was not optimally tracked, which led to the design of an NMPC. Simulation and real-time implementation revealed that the NMPC algorithm results in tight tracking with respect to the nonlinear temperature.

Prajwal Shettigar J et al [4] discusses Wiener-Neural-Network based modelling and validation of Generalized Predictive Control on a laboratory-scale batch reactor. The work methodology deployed was the advanced backpropagation algorithm. This algorithm involves assuming weights for the neural network and then finding the output and the respective error along with the change in the direction of weights in the former step. The WNN-based GPC (WGPC) is

able to track any kind of temperature set point satisfactorily. Hence, the nonlinear tracking of the temperature profile is successfully carried out using the modelling of the batch reactor and designing of the generalized predictive controller.

4. LITERATURE REVIEW SUMMARY

All the reviewed literature papers share a common goal: the efficient production of biodiesel. However, these papers do not specifically address the issue of optimization. Most research in the field of batch reactors focuses on deterministic optimal control problems, yet their applications are based on generic batch reactors and do not consider the kinetics of biodiesel production. In contrast, these research studies have identified the optimal operating conditions that can influence the yield of the transesterification reaction, although their findings are contingent upon the type of feedstock employed. Furthermore, some studies have utilized response surface methodology to investigate the transesterification reaction and have highlighted the key factors influencing the conversion of triglycerides into biodiesel. Nevertheless, these papers primarily concentrate on static optimization, often overlooking the inherent uncertainties associated with biodiesel production.

Biodiesel is typically produced via the transesterification of triglycerides and methanol in the presence of an alkaline catalyst, such as sodium hydroxide. This reaction comprises three consecutive and reversible steps: the conversion of triglycerides (TG) to diglycerides (DG), diglycerides to monoglycerides (MG), and monoglycerides to glycerol. In each step, an ester (RiCOOCH3) is generated, resulting in the production of three ester molecules from one triglyceride molecule.

$$\mathrm{TG} + \mathrm{CH_3OH} \overset{k1,k2}{\Longleftrightarrow} \mathrm{DG} + \mathrm{R_1COOCH_3}$$

$$\mathrm{DG} + \mathrm{CH_3OH} \overset{k3,k4}{\Longleftrightarrow} \mathrm{MG} + \mathrm{R_2COOCH_3}$$

$$\mathrm{MG} + \mathrm{CH_3OH} \overset{k5,k6}{\Longleftrightarrow} \mathrm{GL} + \mathrm{R_3COOCH_3}$$

Overall reaction:

$$\mathrm{TG} + 3\mathrm{CH_3OH} \Longleftrightarrow 3\mathrm{RCOOCH_3} + \mathrm{GL}$$

CHAPTER 3: METHODOLOGY

1. BATCH REACTOR MODEL FOR BIODIESEL

There are 6 major components that are present in biodiesel they are: Triglycerides(TG), Diglycerides(DG), Monoglycerides(MG), Methyl Ester(E), Methanol(A) and Glycerol(GL).

$C_1 = C_{TG} =$ Concentration of Triglycerides
$C_2 = C_{DG} =$ Concentration of Diglycerides
$C_3 = C_{MG} =$ Concentration of Monoglycerides
$C_4 = C_E =$ Concentration of Methyl Ester
$C_5 = C_A =$ Concentration of Methanol
$C_6 = C_{GL} =$ Concentration of Glycerol

The mathematical model for the production of biodiesel in a batch reactor is governed by the following Ordinary Differential Equations (ODEs)(1) from the mass balance of the batch reactor.

$$F_1 = \frac{dC_{TG}}{dt} = -k_1 C_{TG} C_A + k_2 C_{DG} C_E$$
$$F_2 = \frac{dC_{DG}}{dt} = k_1 C_{TG} C_A - k_2 C_{DG} C_E - k_3 C_{DG} C_A + k_4 C_{MG} C_E$$
$$F_3 = \frac{dC_{MG}}{dt} = k_3 C_{DG} C_A - k_4 C_{MG} C_E - k_5 C_{MG} C_A + k_6 C_{GL} C_E$$
$$F_4 = \frac{dC_E}{dt} = k_1 C_{TG} C_A - k_2 C_{DG} C_E + k_3 C_{DG} C_A - k_4 C_{MG} C_E + k_5 C_{MG} C_A - k_6 C_{GL} C_E \qquad \text{……(1)}$$

$$F_5 = \frac{dC_A}{dt} = -k_1 C_{TG} C_A + k_2 C_{DG} C_E - k_3 C_{DG} C_A + k_4 C_{MG} C_E - k_5 C_{MG} C_A + k_6 C_{GL} C_E$$
$$F_6 = \frac{dC_{GL}}{dt} = k_5 C_{MG} C_A - k_6 C_{GL} C_E$$

Where C_{TG}, C_{DG}, C_{MG}, C_E, C_A, C_{GL} are concentrations of triglycerides, diglycerides, monoglycerides, methyl ester, methanol, and glycerol respectively.

The reaction constant k_i (2) is given by:

$$k_i = a_i e^{\frac{-b_i}{T}} \qquad \text{……(2)}$$

Where,
a = frequency factor

$b = E_a/R$
E_a = Activation energy
R = Gas constant
T = Reaction Temperature

2. ITO'S PROCESS

Biodiesel production involves inherent uncertainties that can affect product quality, quantity, and process economics. One of the most common types of uncertainty in biodiesel production is the variability in feed composition.

Stochastic differential equations (SDEs) or Ito processes can be utilized to model the uncertainty present in the feed composition during biodiesel production. SDEs are equations that describe the evolution of a system under uncertainty, such as the variability in the feed composition. Among the various types of SDEs, Ito processes are commonly used to model random processes in several fields, including finance, physics, and engineering. The advantage of utilizing Ito processes is that they can account for the variability in the feed composition and predict the expected concentration of methyl ester in the biodiesel product.

Ito's Lemma, a powerful tool in stochastic calculus, can be used to derive the expected value of a function of a stochastic process. Similar to the chain rule in ordinary calculus, it enables the calculation of the sensitivity of the function to variations in the stochastic process. In biodiesel production, Ito's Lemma can be employed to estimate the expected concentration of methyl ester in the biodiesel product, accounting for the uncertainty in the feed composition.

A general Ito's Process is defined by: $dx = a(x,t)dt + b(x,t)dz$

For the study in this project, a simplification of the above equation is used to represent time dependent uncertainties in the concentration for each component, known as the Brownian motion with drift.

Given by: $dC_i = F_i(C,t)\Delta t + g_i \varepsilon_t \sqrt{\Delta t}$

Where, C_i represents the state variable, which is the concentration for each component. F_i (c, t) is the right-hand side of the differential equation for the concentration of components i and g_i(3), which are the variance parameters. Over any time interval dt, the change in C_i is normally distributed and the value of g_i (3) can be calculated by computing the square root of the variance of the differences in C_i and divided by the interval time dt.

$$g_i = \sqrt{\frac{\mathrm{var}(\Delta C_i)}{\Delta t}} \qquad \text{……(3)}$$

Concentration profiles for each component of biodiesel production before and after applying Ito's Process:

The following plots (Fig.2.1-2.12) shows the concentration of the 6 major elements of biodiesel before and after the application of Ito's Process on MS Excel.

I. Concentration of Triglycerides:

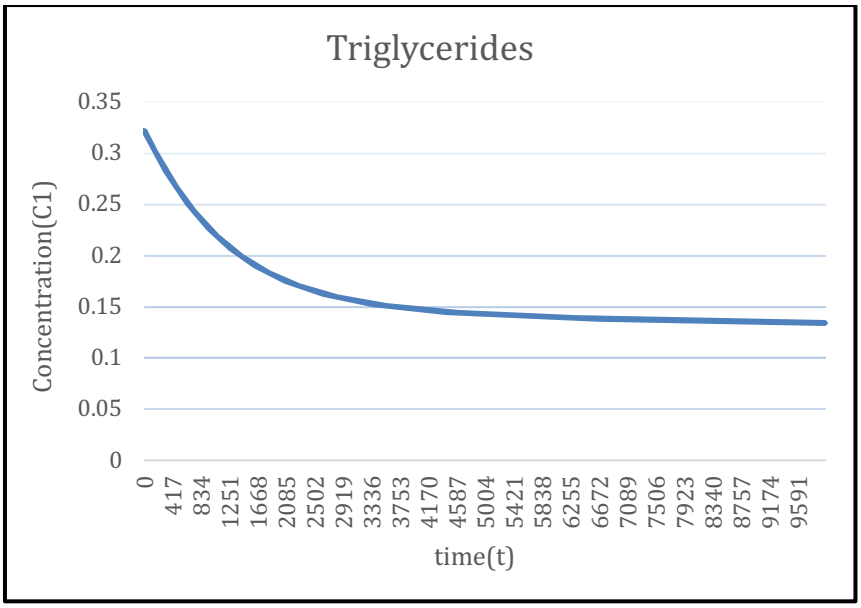

Fig. 2.1 Concentration of Triglycerides before applying Ito Process

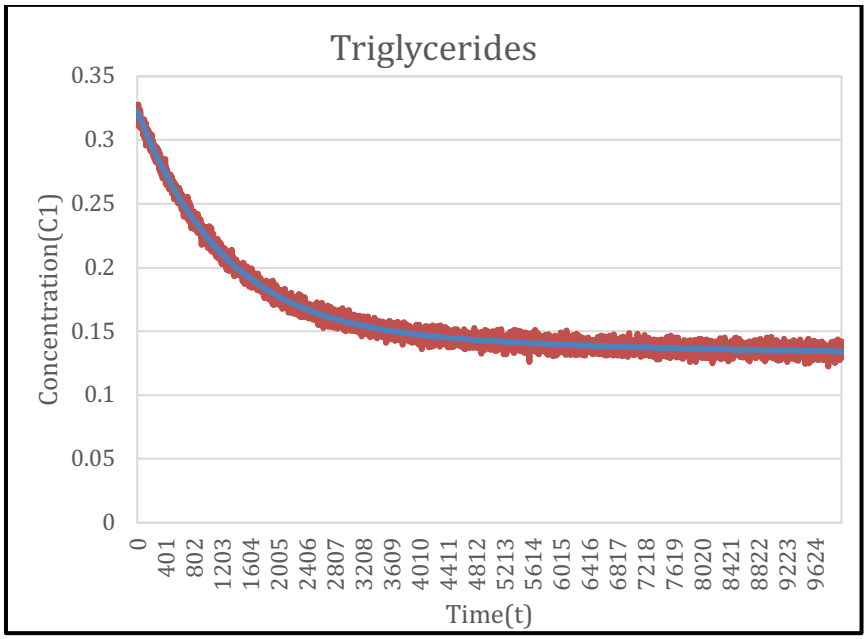

Fig. 2.2 Concentration of Triglycerides after applying Ito Process

II. Concentration of Diglycerides:

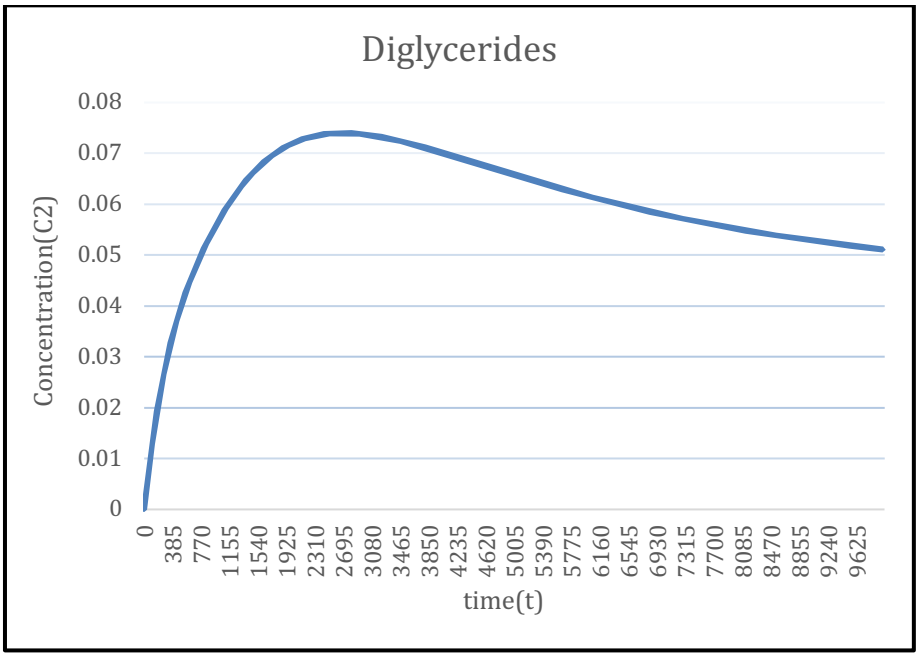

Fig. 2.3 Concentration of Diglycerides before applying Ito Process

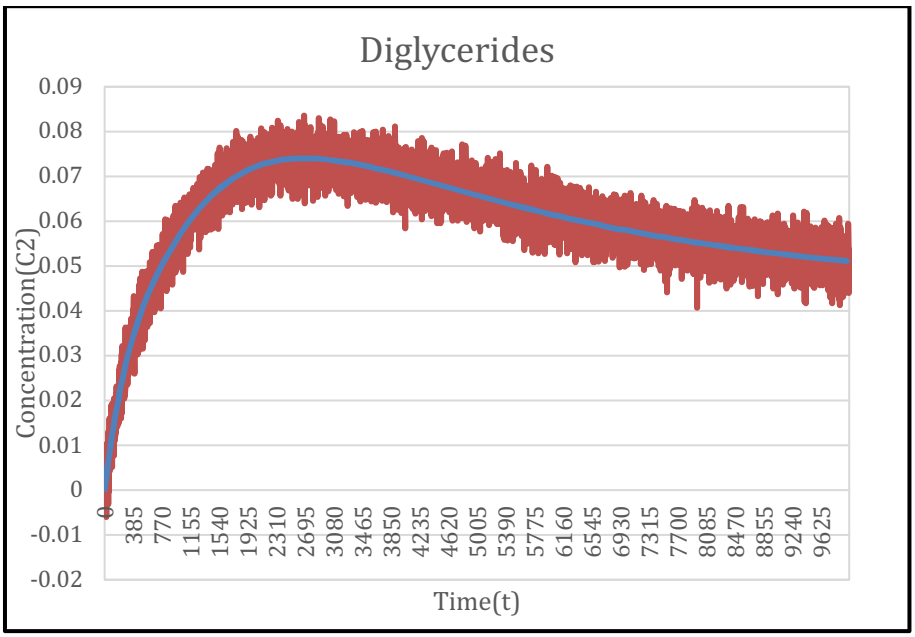

Fig. 2.4 Concentration of Diglycerides after applying Ito Process

III. Concentration of Monoglycerides

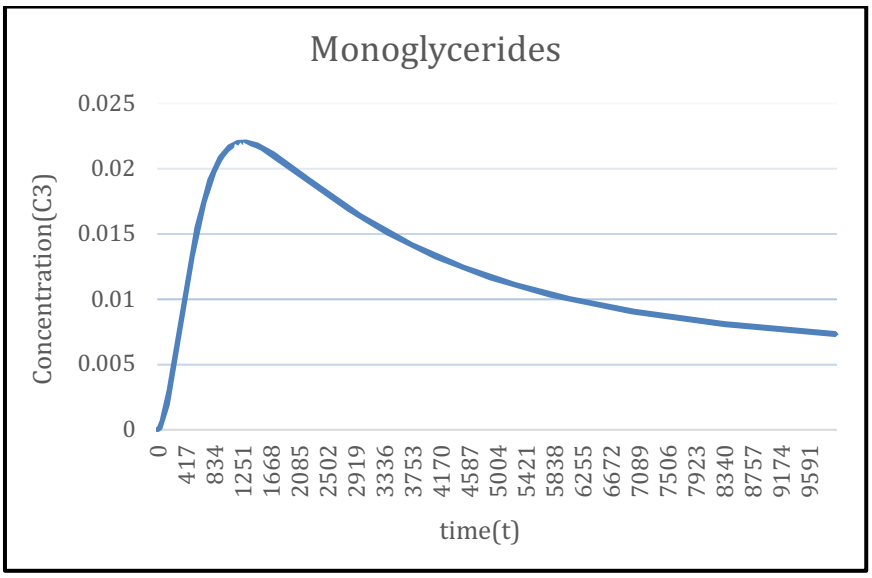

Fig. 2.5 Concentration of Monoglycerides before applying Ito Process

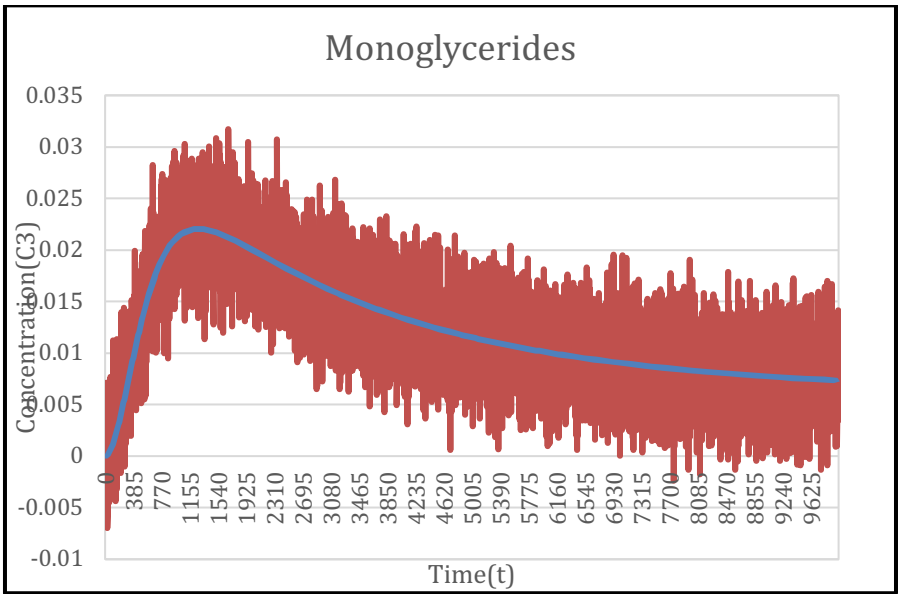

Fig. 2.6 Concentration of Monoglycerides after applying Ito Process

IV. Concentration of Methyl Ester

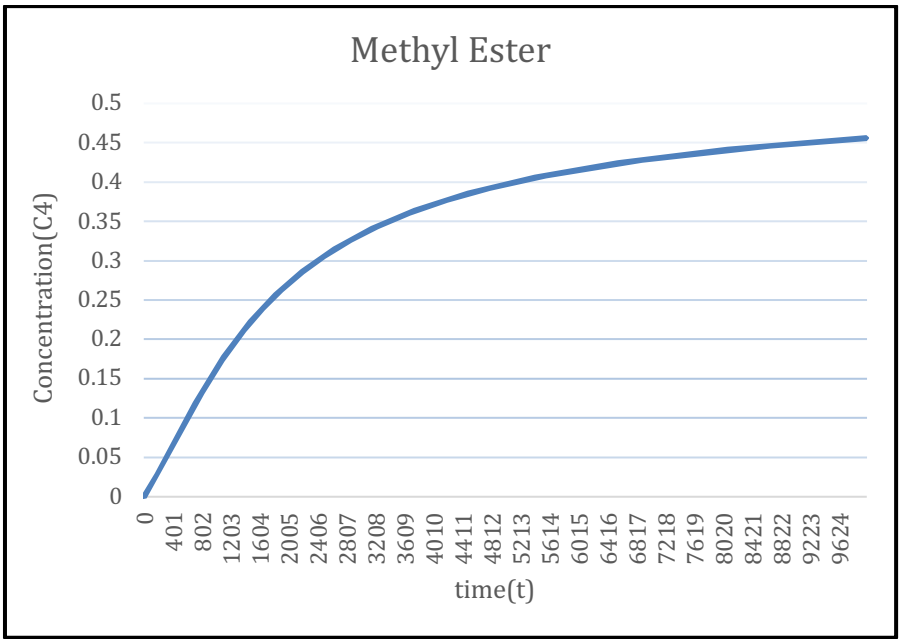

Fig.2.7 Concentration of Methyl Ester before applying Ito Process

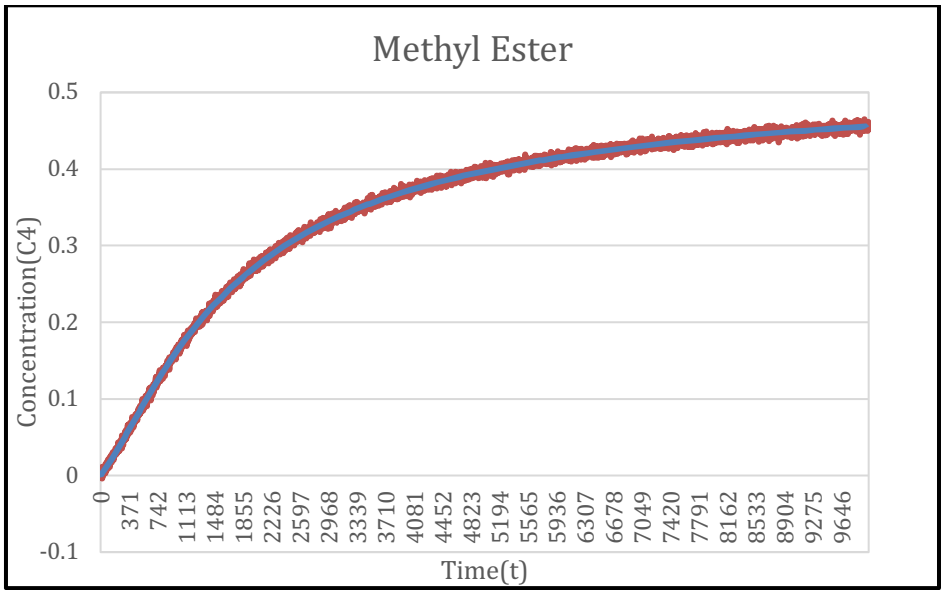

Fig. 2.8 Concentration of Methyl Ester after applying Ito Process

V. Concentration of Methanol

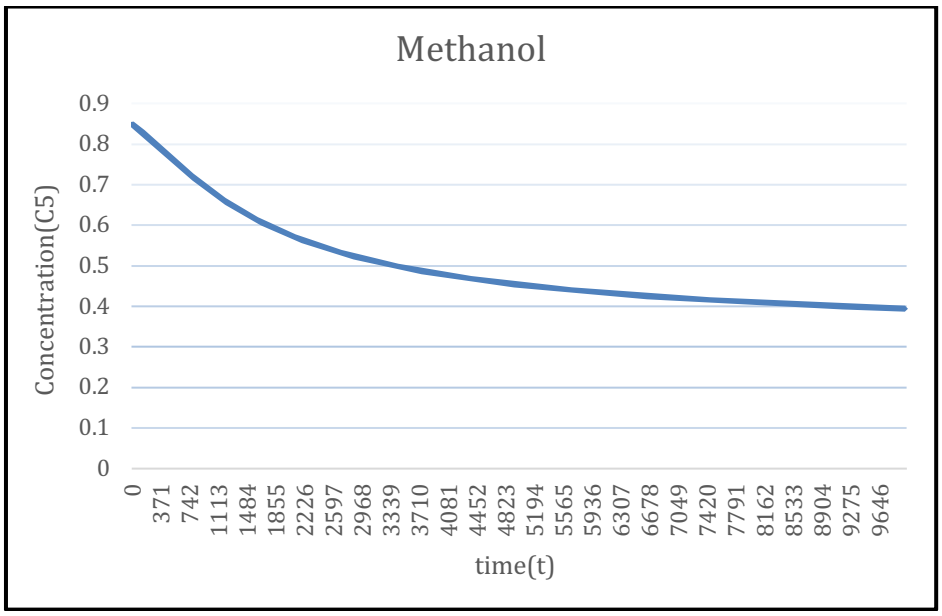

Fig. 2.9 Concentration of Methanol before applying Ito Process

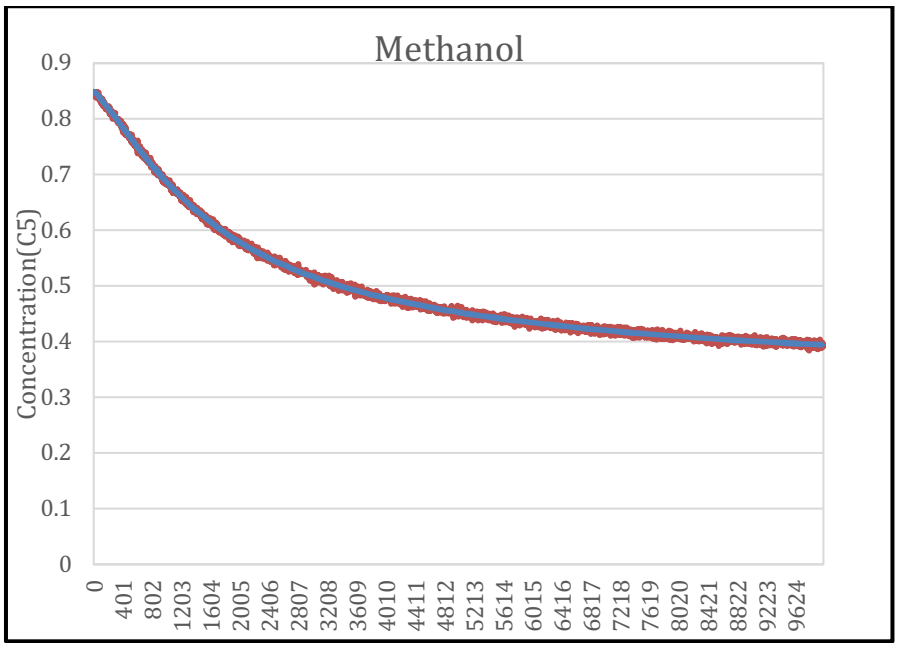

Fig. 2.10 Concentration of Methanol after applying Ito Process

VI. Concentration of Glycerol

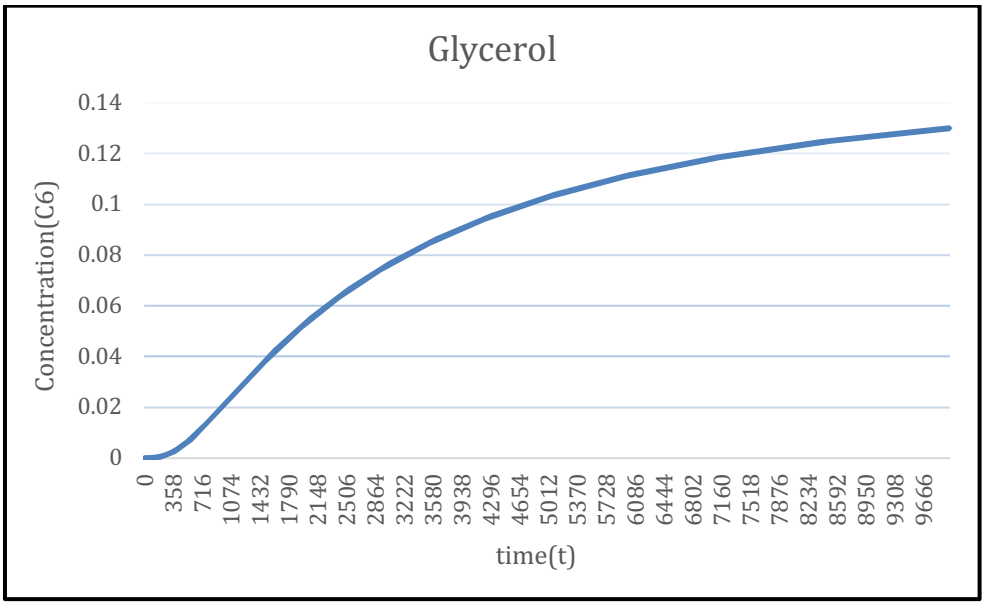

Fig. 2.11 Concentration of Glycerol before applying Ito Process

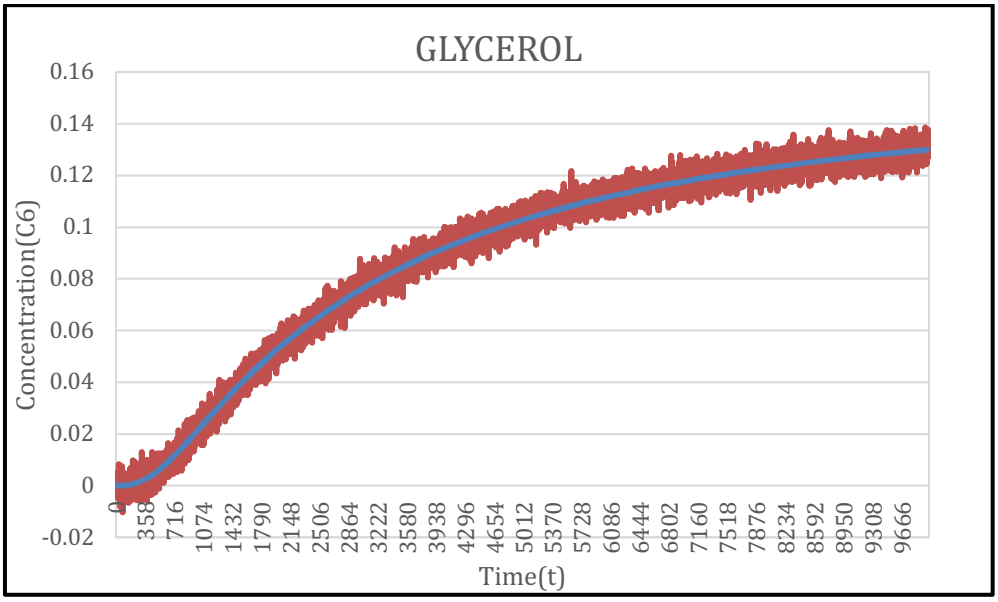

Fig. 2.12 Concentration of Glycerol after applying Ito Process

3. OPTIMAL CONTROL PROBLEM

I. Stochastic Maximum Principle

The stochastic maximum principle (SMP) is a principle in mathematical optimization that provides necessary conditions for finding optimal solutions to stochastic control problems. Stochastic control problems involve making decisions under uncertainty, where the system being controlled is subject to random disturbances. In our case, these disturbances are uncertainties in feedstock composition of biodiesel.

This principle can be used to determine the optimal control policy that maximizes the expected value of the objective function over a given time period.

In this project, we will be maximising the expected value of methyl ester (biodiesel) i.e., equation (4), by finding the best temperature profile in a given period of time. For this study, our time period is 100 minutes.

The objective function is given by:

$$\max L = E\left[\int_{t_0}^{t_f} k_1 C_{TG} C_A - k_2 C_{DG} C_E + k_3 C_{DG} C_A - k_4 C_{MG} C_E + k_5 C_{MG} C_A - k_6 C_{GL} C_E\right] dt \qquad \text{.....(4)}$$

Rewriting the above equation,

$$\max L = E\left[C_E(t_f)\right]$$

Where, E[ ] is the expected value.

Subject the equation to an Ito Process we get,

$$dC_i = F_i(C,t)\Delta t + g_i \varepsilon_t \sqrt{\Delta t}$$

Where, g_i is the variance parameter.

The corresponding optimality condition is given by:

$$0 = \max_{Tf}\left[\frac{1}{dt} E(dL)\right]$$

II. Ito's Lemma

To work with stochastic processes, we make use of Ito's lemma, which allows us to differentiate and integrate functions of the stochastic processes. Using this Ito's lemma we obtain equation (6) and is called the fundamental stochastic calculus theorem. It is the stochastic calculus counterpart of the chain rule in ordinary calculus.

Thus, applying the Ito's lemma to optimality condition equation (4),

$$0 = \frac{\partial L}{\partial t} + \sum_{i=1}^{6} \frac{\partial L}{\partial C_i} F_i(C_t, t_t) + \sum_{i=1}^{6} \frac{g_j}{2} \frac{\partial^2 L}{\partial C_i^{\,2}} + \sum_{i=1}^{6} g_i g_j \frac{\partial^2 L}{\partial C_i \partial C_j} \qquad \text{……. (6)}$$

$$0 = \frac{\partial L}{\partial t} + Max \left[\begin{array}{l} \frac{\partial L}{\partial C_{TG}} F_1 + \frac{\partial L}{\partial C_{DG}} F_2 + \frac{\partial L}{\partial C_{MG}} F_3 + \frac{\partial L}{\partial C_E} F_4 + \frac{\partial L}{\partial C_A} F_5 + \frac{\partial L}{\partial C_{GL}} F_6 + \frac{g_{TG}^2}{2} \frac{\partial^2 L}{\partial C_{TG}^2} \\ + \frac{g_{DG}^2}{2} \frac{\partial^2 L}{\partial C_{DG}^2} + \frac{g_{MG}^2}{2} \frac{\partial^2 L}{\partial C_{MG}^2} + \frac{g_E^2}{2} \frac{\partial^2 L}{\partial C_E^2} + \frac{g_A^2}{2} \frac{\partial^2 L}{\partial C_A^2} + \frac{g_{GL}^2}{2} \frac{\partial^2 L}{\partial C_{GL}^2} \end{array} \right]$$

Where, z_i and ω_i are adjoint variables for triglycerides, diglycerides, monoglycerides, methyl ester, methanol, and glycerol.

The adjoint equations (7) and (8) to be solved in the stochastic maximum principle formulation are:

$$\frac{dz_j}{dt} = \sum_{i=1}^{6} \left[-\frac{\partial F_i}{\partial C_j} Z_i - \frac{1}{2} \left(\frac{\partial g_i^2}{\partial C_j} \right) \omega_i \right] \qquad \text{……(7)}$$

$$\frac{d\omega_j}{dt} = \sum_{i=1}^{6} \left[-2\omega_i \frac{\partial F_i}{\partial C_j} - z_i \frac{\partial^2 F_i}{\partial C_j^{\,2}} - \frac{1}{2} \left(\frac{\partial^2 (g_i^{\,2})}{\partial C_j^{\,2}} \right) \omega_i \right] \qquad \text{……(8)}$$

III. The Hamiltonian

The maximum principle formulation results in two point boundary value problem, where the initial conditions for the state variables Ci are known, but the conditions for the adjoint variables are only known at the final boundary.

This algorithm starts with the initial estimate temperature T(t). The equations are solved by employing the RKF method with a step size equal to 0.01 to obtain the Hamiltonian(H) (9).

Next, the values of dH/dT at each time are computed and then the convergence criterion (dH/dT < tolerance) is verified. If the convergence criterion is not satisfied, the temperature T(t) is updated using this gradient, in such that the updated temperature profile improves the objective function.

$$H = Z_1F_1 + Z_2F_2 + Z_3F_3 + Z_4F_4 + Z_5F_5 + Z_6F_6 + \frac{g_{TG}^2}{2}\omega_1 + \frac{g_{DG}^2}{2}\omega_2 + \frac{g_{MG}^2}{2}\omega_3 + \frac{g_E^2}{2}\omega_4 + \frac{g_A^2}{2}\omega_5 + \frac{g_{GL}^2}{2}\omega_6 \quad \text{……. (9)}$$

The optimal decision vector T(t) can be obtained by finding the extremum of the Hamiltonian at each time step. That is applying the optimality condition dH/dT < Tolerance. Where, M is a suitable constant that can be small enough so that no instability will result or large enough for rapid convergence.

$$T(t)^{new} = T(t)^{old} + M\left(\frac{dH}{dT}(t)\right)$$

An analytical approach is used in order to calculate the derivative of the Hamiltonian (9) to determine the optimal temperature trajectory for this study. The following derivates were calculated with the help of the following partial differential equations:

Partial Differential Equations used to solve the adjoint equations z and ω

1.
$$\frac{\partial F_1}{\partial C_1} = -k_1 C_A \qquad \frac{\partial F_2}{\partial C_1} = k_1 C_A \qquad \frac{\partial F_3}{\partial C_1} = 0 \qquad \frac{\partial F_4}{\partial C_1} = k_1 C_A \qquad \frac{\partial F_5}{\partial C_1} = 0 \qquad \frac{\partial F_6}{\partial C_1} = 0$$

2.
$$\frac{\partial F_1}{\partial C_2} = k_2 C_E \qquad \frac{\partial F_2}{\partial C_2} = -k_2 C_E - k_3 C_A \qquad \frac{\partial F_3}{\partial C_2} = k_3 C_A \qquad \frac{\partial F_4}{\partial C_2} = -k_2 C_E + k_3 C_A \qquad \frac{\partial F_5}{\partial C_2} = 0 \qquad \frac{\partial F_6}{\partial C_2} = 0$$

3.
$$\frac{\partial F_1}{\partial C_3} = 0 \qquad \frac{\partial F_2}{\partial C_3} = k_4 C_E \qquad \frac{\partial F_3}{\partial C_3} = -k_4 C_E - k_5 C_A \qquad \frac{\partial F_4}{\partial C_3} = -k_4 C_E + k_5 C_A \qquad \frac{\partial F_5}{\partial C_3} = 0 \qquad \frac{\partial F_6}{\partial C_3} = k_5 C_A$$

4.
$$\frac{\partial F_1}{\partial C_4} = k_2 C_{DG} \qquad \frac{\partial F_2}{\partial C_4} = -k_2 C_{DG} + k_4 C_{MG} \qquad \frac{\partial F_3}{\partial C_4} = -k_4 C_{MG} + k_6 C_{GL} \qquad \frac{\partial F_4}{\partial C_4} = -k_2 C_{DG} - k_4 C_{MG} - k_6 C_{GL} \qquad \frac{\partial F_5}{\partial C_4} = k_2 C_{DG} + k_4 C_{MG} + k_6 C_{GL} \qquad \frac{\partial F_6}{\partial C_4} = -k_6 C_{GL}$$

5.
$$\frac{\partial F_1}{\partial C_5} = -k_1 C_{TG} \qquad \frac{\partial F_2}{\partial C_5} = k_1 C_{TG} - k_3 C_{DG} \qquad \frac{\partial F_3}{\partial C_5} = k_3 C_{DG} - k_5 C_{MG} \qquad \frac{\partial F_4}{\partial C_5} = k_1 C_{TG} + k_3 C_{DG} + k_5 C_{MG} \qquad \frac{\partial F_5}{\partial C_5} = -k_1 C_{TG} - k_3 C_{DG} - k_5 C_{MG} \qquad \frac{\partial F_6}{\partial C_5} = k_5 C_{MG}$$

6.
$$\frac{\partial F_1}{\partial C_6} = 0 \qquad \frac{\partial F_2}{\partial C_6} = 0 \qquad \frac{\partial F_3}{\partial C_6} = k_6 C_E \qquad \frac{\partial F_4}{\partial C_6} = -k_6 C_E \qquad \frac{\partial F_5}{\partial C_6} = 0 \qquad \frac{\partial F_6}{\partial C_6} = -k_6 C_E$$

a. Adjoint equation for z

As mentioned in Equation (7),

$$\frac{dz_j}{dt} = \sum_{i=1}^{6} \left[-\frac{\partial F_i}{\partial C_j} z_i - \frac{1}{2}\left(\frac{\partial g_i{}^2}{\partial C_j{}^2}\right)\omega_i \right]$$

For j=1,

$$dz_1 = \left[-\frac{\partial F_1}{\partial C_1} z_1 - \frac{1}{2}\left(\frac{\partial g_1{}^2}{\partial C_1{}^2}\right)\omega_1\right] + \left[-\frac{\partial F_2}{\partial C_1} z_2 - \frac{1}{2}\left(\frac{\partial g_2{}^2}{\partial C_1{}^2}\right)\omega_2\right] + \left[-\frac{\partial F_3}{\partial C_1} z_3 - \frac{1}{2}\left(\frac{\partial g_3{}^2}{\partial C_1{}^2}\right)\omega_3\right]$$
$$+\left[-\frac{\partial F_4}{\partial C_1} z_4 - \frac{1}{2}\left(\frac{\partial g_4{}^2}{\partial C_1{}^2}\right)\omega_4\right] + \left[-\frac{\partial F_5}{\partial C_1} z_5 - \frac{1}{2}\left(\frac{\partial g_5{}^2}{\partial C_1{}^2}\right)\omega_5\right] + \left[-\frac{\partial F_6}{\partial C_1} z_6 - \frac{1}{2}\left(\frac{\partial g_6{}^2}{\partial C_1{}^2}\right)\omega_6\right]$$

Similarly for j = 2 … 6, we can derive $\frac{dz_2}{dt}$ to $\frac{dz_6}{dt}$

$$\frac{dz_1}{dt} = -z_1(-k_1C_A) - z_2(k_1C_A) - z_4(k_1C_A) - z_5(-k_1C_A)$$
$$\frac{dz_2}{dt} = -z_1(k_2C_E) - z_2(-k_2C_E - k_3C_A) - z_3(k_3C_A) - z_4(-k_2C_E + k_3C_A) - z_5(k_2C_E - k_3C_A)$$
$$\frac{dz_3}{dt} = -z_2(k_4C_E) - z_3(-k_4C_E - k_5C_A) - z_4(-k_4C_E + k_5C_A) - z_5(k_4C_E - k_5C_A) - z_6(k_5C_A)$$
$$\frac{dz_4}{dt} = -z_1(k_2C_{DG}) - z_2(-k_2C_{DG} + k_4C_{MG}) - z_3(-k_4C_{MG} + k_6C_{GL}) - z_4(-k_2C_{DG} - k_4C_{MG} - k_6C_{GL}) - z_5(k_2C_{DG} + k_4C_{MG} + k_6C_{GL}) + z_6(k_6C_{Gl})$$
$$\frac{dz_5}{dt} = z_1(k_1C_{TG}) - z_2(k_1C_{TG} - k_3C_{DG}) - z_3(k_3C_{DG} - k_5C_{MG}) - z_4(k_1C_{TG} + k_3C_{DG} + k_5C_{MG}) - z_5(-k_1C_{TG} - k_3C_{DG} - k_5C_{MG}) - z_6(k_5C_{MG})$$
$$\frac{dz_6}{dt} = -z_3(k_6C_E) - z_4(k_6C_E) - z_5(k_6C_E) + z_6(k_6C_E)$$

b. <u>Adjoint Equation for ω</u>

As mentioned in Equation (8),

$$\frac{d\omega_j}{dt} = \sum_{i=1}^{6}\left[-\left(\frac{2\omega_i \partial F_i}{\partial C_j}\right) - \left(\frac{z_i \partial^2 F_i}{\partial C^2{}_j}\right) - \frac{1}{2}\left(\frac{\partial^2 g_i{}^2}{\partial C^2{}_j}\right)\right]$$

$$\frac{d\omega_1}{dt} = \left[-\left(\frac{2\omega_1 \partial F_1}{\partial C_1}\right) - \left(\frac{z_1 \partial^2 F_1}{\partial C_1^2}\right) - \frac{1}{2}\left(\frac{\partial^2 g_1{}^2}{\partial C_1^2}\right)\omega_1\right] + \left[-\left(\frac{2\omega_2 \partial F_2}{\partial C_1}\right) - \left(\frac{z_2 \partial^2 F_2}{\partial C_1^2}\right) - \frac{1}{2}\left(\frac{\partial^2 g_2{}^2}{\partial C_1^2}\right)\omega_2\right] + \left[-\left(\frac{2\omega_3 \partial F_3}{\partial C_1}\right) - \left(\frac{z_3 \partial^2 F_3}{\partial C_1^2}\right) - \frac{1}{2}\left(\frac{\partial^2 g_3{}^2}{\partial C_1^2}\right)\omega_3\right]$$
$$+\left[-\left(\frac{2\omega_4 \partial F_4}{\partial C_1}\right) - \left(\frac{z_4 \partial^2 F_4}{\partial C_1^2}\right) - \frac{1}{2}\left(\frac{\partial^2 g_4{}^2}{\partial C_1^2}\right)\omega_4\right] + \left[-\left(\frac{2\omega_5 \partial F_5}{\partial C_1}\right) - \left(\frac{z_5 \partial^2 F_5}{\partial C_1^2}\right) - \frac{1}{2}\left(\frac{\partial^2 g_5{}^2}{\partial C_1^2}\right)\omega_5\right] + \left[-\left(\frac{2\omega_6 \partial F_6}{\partial C_1}\right) - \left(\frac{z_6 \partial^2 F_6}{\partial C_1^2}\right) - \frac{1}{2}\left(\frac{\partial^2 g_6{}^2}{\partial C_1^2}\right)\omega_6\right]$$

Similarly for j = 2 … 6, we can derive $\frac{d\omega_2}{dt}$ to $\frac{d\omega_6}{dt}$

$$\frac{d\omega_1}{dt} = 2\omega_1 k_1 C_5 - 2\omega_2 k_1 C_5 - 2\omega_4 k_1 C_5 + 2\omega_5 k_1 C_5$$

$$\frac{d\omega_2}{dt} = -2\omega_1 k_2 C_4 + 2\omega_2 k_2 C_4 + 2\omega_2 k_3 C_5 - 2\omega_3 k_3 C_5 + 2\omega_4 k_2 C_4 - 2\omega_4 k_3 C_5$$

$$\frac{d\omega_3}{dt} = -2\omega_2 k_4 C_4 + 2\omega_3 k_4 C_4 + 2\omega_3 k_5 C_5 + 2\omega_4 k_4 C_4 - 2\omega_4 k_5 C_5 - 2\omega_6 k_5 C_5$$

$$\frac{d\omega_4}{dt} = -2\omega_1 k_2 C_2 + 2\omega_2 k_2 C_2 - 2\omega_2 k_4 C_1 + 2\omega_3 k_4 C_1 - 2\omega_3 k_6 C_6 + 2\omega_4 k_2 C_2 + 2\omega_4 k_4 C_1 + 2\omega_4 k_6 C_6 - 2\omega_5 k_2 C_2 - 2\omega_5 k_4 C_1 - 2\omega_5 k_6 C_6 + 2\omega_6 k_6 C_6$$

$$\frac{d\omega_5}{dt} = 2\omega_1 k_1 C_3 - 2\omega_2 k_1 C_3 + 2\omega_2 k_3 C_2 - 2\omega_3 k_3 C_2 + 2\omega_3 k_5 C_1 - 2\omega_4 k_1 C_3 - 2\omega_4 k_3 C_2 - 2\omega_4 k_5 C_1 + 2\omega_5 k_1 C_3 + 2\omega_5 k_3 C_2 + 2\omega_5 k_5 C_1 - 2\omega_6 k_5 C_1$$

$$\frac{d\omega_6}{dt} = -2\omega_3 k_6 C_4 + 2\omega_4 k_6 C_4 - 2\omega_5 k_6 C_4 + 2\omega_6 k_6 C_4$$

The initial conditions for the adjoint variables are set as z_i (tf) = [0;0;0;1;0;0] and ω_i(tf) = [0;0;0;0;0;0] representing the fixed values of vector A. Employing backward integration in conjunction with the RKF method enables us to solve these equations. Ultimately, the optimal decision vector T(t) can be derived by identifying the extremum of the Hamiltonian at each time increment.

As previously mentioned, the maximum principle plays a pivotal role in optimal control theory when seeking the most favourable profile for a dynamic system. This approach underscores the necessity of maximizing the Hamiltonian concerning the control variable, essentially applying the optimality condition dH/dT < tolerance.

In the following section, we delve into the derivation of the Hamiltonian's derivative to ascertain the optimal temperature trajectory for this particular problem. This challenge demands the utilization of the total derivative concept, which asserts that when a continuous function, denoted as z = f(x, y), relies on multiple variables like t, x, y, and so forth, and is differentiated concerning one of these input variables, the derivative can be expressed as a series of partial derivatives. This results in the ability to compute the rate of change of 'z' and 'ω' with respect to time(t) as follows:

$$\frac{dz}{dt} = \frac{\delta z}{\delta x}\left(\frac{dx}{dt}\right) + \frac{\delta z}{\delta y}\left(\frac{dy}{dt}\right)$$

$$\frac{d\omega}{dt} = \frac{\delta \omega}{\delta x}\left(\frac{dx}{dt}\right) + \frac{\delta \omega}{\delta y}\left(\frac{dy}{dt}\right)$$

Applying the total derivative, we can calculate analytically the derivative of the Hamiltonian (10) as:

$$\frac{dH}{dT} = \sum_{i=1}^{6} \frac{\delta H}{\delta C_i}\left(\frac{dC_i}{dt}\right) + \sum_{i=1}^{6} \frac{\delta H}{\delta z_i}\left(\frac{dz_i}{dt}\right) + \sum_{i=1}^{6} \frac{\delta H}{\delta \omega_i}\left(\frac{d\omega_i}{dt}\right) \quad \text{…….(10)}$$

With the following boundary conditions: $z_i(tf) = [0;0;0;1;0;0]$ and $\omega_i = [0;0;0;0;0;0]$.

Considering the following expressions:

$$\theta_i = \frac{dC_i}{dt} \quad \text{.....(11)}$$

$$\phi_i = \frac{dz_i}{dt} \quad \text{……(12)}$$

$$\psi_i = \frac{d\omega_i}{dt} \quad \text{……(13)}$$

Substituting the above equations (11), (12), (13) in Hamiltonian derivative (10) , we get:

$$\frac{dH}{dT} = \sum_{i=1}^{6} \frac{\delta H}{\delta C_i}\theta_i + \sum_{i=1}^{6} \frac{\delta H}{\delta z_i}\phi_i + \sum_{i=1}^{6} \frac{\delta H}{\delta \omega_i}\psi_i \quad \text{…….(14)}$$

Applying the following mathematical property for each component:

$$\frac{d}{dT}\left(\frac{dC_i}{dt}\right) = \frac{d}{dt}\left(\frac{dC_i}{dT}\right) = \frac{d\theta_i}{dt} \quad \text{……(15)}$$

$$\frac{d}{dT}\left(\frac{dz_i}{dt}\right) = \frac{d}{dt}\left(\frac{dz_i}{dT}\right) = \frac{d\phi_i}{dt} \quad \text{……(16)}$$

$$\frac{d}{dT}\left(\frac{d\omega_i}{dt}\right) = \frac{d}{dt}\left(\frac{d\omega_i}{dT}\right) = \frac{d\psi_i}{dt} \quad \text{……(17)}$$

With the following boundary conditions: $\theta_i = [0;0;0;0;0;0]$, $\Phi_i = [0;0;0;0;0;0]$ and $\Psi_i = [0;0;0;0;0;0]$

Finally, we obtain 6 differential equations for each of the expressions $d\theta_i/dt$ (15), $d\Phi_i/dt$ (16) and $d\Psi_i/dt$ (17). We then use Runge Kutta Fehlberg (RKF) method to solve the Hamiltonian (14) equation using integration. Forward integration is used on $d\theta_i/dt$ to obtain θ_i and backward integration is used to solve $d\Phi_i/dt$ and $d\Psi_i/dt$ to obtain Φ_i and Ψ_i respectively.

The 6 derivative equations for each of $d\theta_i/dt$(18), $d\Phi_i/dt$(19) and $d\Psi_i/dt$(20) are as follows:

a) <u>Theta Equations:</u>

$$\frac{d\theta_1}{dt} = -\left(\frac{dk_1}{dT}C_{TG}C_A + k_1\theta_{TG}C_A + k_1C_{TG}\theta_A\right) + \left(\frac{dk_2}{dT}C_{DG}C_E + k_2\theta_{DG}C_E + k_2C_{DG}\theta_E\right)$$

$$\begin{aligned}\frac{d\theta_2}{dt} = &\left(\frac{dk_1}{dT}C_{TG}C_A + k_1\theta_{TG}C_A + k_1C_{TG}\theta_A\right) - \left(\frac{dk_2}{dT}C_{DG}C_E + k_2\theta_{DG}C_E + k_2C_{DG}\theta_E\right)\\ &- \left(\frac{dk_3}{dT}C_{DG}C_A + k_3\theta_{DG}C_A + k_3C_{DG}\theta_A\right) + \left(\frac{dk_4}{dT}C_{MG}C_E + k_4\theta_{MG}C_E + k_4C_{MG}\theta_E\right)\end{aligned}$$

$$\begin{aligned}\frac{d\theta_3}{dt} = &\left(\frac{dk_3}{dT}C_{DG}C_A + k_3\theta_{DG}C_A + k_3C_{DG}\theta_A\right) - \left(\frac{dk_4}{dT}C_{MG}C_E + k_4\theta_{MG}C_E + k_4C_{MG}\theta_E\right)\\ &- \left(\frac{dk_5}{dT}C_{MG}C_A + k_5\theta_{MG}C_A + k_5C_{MG}\theta_A\right) + \left(\frac{dk_6}{dT}C_{GL}C_E + k_6\theta_{GL}C_E + k_6C_{GL}\theta_E\right)\end{aligned}$$

$$\begin{aligned}\frac{d\theta_4}{dt} = &\left(\frac{dk_1}{dT}C_{TG}C_A + k_1\theta_{TG}C_A + k_1C_{TG}\theta_A\right) - \left(\frac{dk_2}{dT}C_{DG}C_E + k_2\theta_{DG}C_E + k_2C_{DG}\theta_E\right)\\ &+ \left(\frac{dk_3}{dT}C_{DG}C_A + k_3\theta_{DG}C_A + k_3C_{DG}\theta_A\right) - \left(\frac{dk_4}{dT}C_{MG}C_E + k_4\theta_{MG}C_E + k_4C_{MG}\theta_E\right)\\ &+ \left(\frac{dk_5}{dT}C_{MG}C_A + k_5\theta_{MG}C_A + k_5C_{MG}\theta_A\right) - \left(\frac{dk_6}{dT}C_{GL}C_E + k_6\theta_{GL}C_E + k_6C_{GL}\theta_E\right)\end{aligned}$$

$$\begin{aligned}\frac{d\theta_5}{dt} = &-\left(\frac{dk_1}{dT}C_{TG}C_A + k_1\theta_{TG}C_A + k_1C_{TG}\theta_A\right) + \left(\frac{dk_2}{dT}C_{DG}C_E + k_2\theta_{DG}C_E + k_2C_{DG}\theta_E\right)\\ &- \left(\frac{dk_3}{dT}C_{DG}C_A + k_3\theta_{DG}C_A + k_3C_{DG}\theta_A\right) + \left(\frac{dk_4}{dT}C_{MG}C_E + k_4\theta_{MG}C_E + k_4C_{MG}\theta_E\right)\\ &- \left(\frac{dk_5}{dT}C_{MG}C_A + k_5\theta_{MG}C_A + k_5C_{MG}\theta_A\right) + \left(\frac{dk_6}{dT}C_{GL}C_E + k_6\theta_{GL}C_E + k_6C_{GL}\theta_E\right)\end{aligned}$$

$$\frac{d\theta_6}{dt} = \left(\frac{dk_5}{dT}C_{MG}C_A + k_5\theta_{MG}C_A + k_5C_{MG}\theta_A\right) - \left(\frac{dk_6}{dT}C_{GL}C_E + k_6\theta_{GL}C_E + k_6C_{GL}\theta_E\right)$$

……(18)

b) <u>Phi Equations:</u>

$$\frac{d\phi_1}{dt} = \phi_{TG}k_1C_A + z_1\frac{dk_1}{dT}C_A + z_1k_1\theta_A - \phi_{DG}k_1C_A - z_2\frac{dk_1}{dT}C_A - z_2k_1\theta_A - \phi_Ek_1C_A - z_4\frac{dk_1}{dT}C_A - z_4k_1\theta_A + \phi_Ak_1C_A + z_5\frac{dk_1}{dT}C_A + z_5k_1\theta_A$$

$$\frac{d\phi_2}{dt} = -\phi_{TG}k_2C_A - z_1\frac{dk_2}{dT}C_A - z_1k_2\theta_A + \phi_{DG}k_2C_E + z_2\frac{dk_2}{dT}C_E - z_2k_2\theta_E + \phi_{DG}k_3C_A + z_2\frac{dk_3}{dT}C_A + z_2k_3\theta_A - \phi_{MG}k_3C_A - z_3\frac{dk_3}{dT}C_A - z_3k_3\theta_A + \phi_Ek_2C_E + z_4\frac{dk_2}{dT}C_E + z_4k_2\theta_E + \phi_Ek_3C_A + z_4\frac{dk_3}{dT}C_A + z_4k_3\theta_A - \phi_Ak_2C_E - z_5\frac{dk_2}{dT}C_E - z_5k_2\theta_A + \phi_Ak_3C_A + z_5\frac{dk_3}{dT}C_A + z_5k_3\theta_A$$

$$\frac{d\phi_3}{dt} = -\phi_{DG}k_4C_E - z_2\frac{dk_4}{dT}C_E - z_2k_4\theta_E + \phi_{MG}k_4C_E + z_4\frac{dk_4}{dT}C_E + z_3k_4\theta_E + \phi_{MG}k_5C_A + z_3\frac{dk_5}{dT}C_A + z_3k_5\theta_A + \phi_Ek_4C_E + z_4\frac{dk_4}{dT}C_E + z_4k_4\theta_E - \phi_Ek_5C_A - z_4\frac{dk_5}{dT}C_A - z_4k_5\theta_A - \phi_Ak_4C_E - z_5\frac{dk_4}{dT}C_E - z_5k_4\theta_E + \phi_Ak_5C_A + z_5\frac{dk_5}{dT}C_A + z_5k_5\theta_A - \phi_{GL}k_5C_A - z_6\frac{dk_5}{dT}C_A - z_6k_5\theta_A$$

$$\frac{d\phi_4}{dt} = -\phi_{TG}k_2C_{DG} - z_1\frac{dk_2}{dT}C_{DG} - z_1k_2\theta_{DG} + \phi_{DG}k_2C_{DG} + z_2\frac{dk_2}{dT}C_{DG} + z_2k_2\theta_{DG} - \phi_{MG}k_4C_{MG} - z_2\frac{dk_4}{dT}C_{MG} - z_2k_4\theta_{MG} + \phi_{MG}k_4C_{MG} + z_3\frac{dk_4}{dT}C_{MG} + z_3k_4\theta_{MG} - \phi_{MG}k_6C_{GL} - z_3\frac{dk_6}{dT}C_{GL} - z_3k_6\theta_{GL} + \phi_Ek_2C_{DG} + z_4\frac{dk_2}{dT}C_{DG} + z_4k_2\theta_{DG} + \phi_Ek_4C_{MG} + z_4\frac{dk_4}{dT}C_{MG} + z_4k_4\theta_{MG} + \phi_Ek_6C_{GL} + z_4\frac{dk_6}{dT}C_{GL} + z_4k_6\theta_{GL} - \phi_Ak_2C_{DG} - z_5\frac{dk_2}{dT}C_{DG} - z_5k_2\theta_{DG} - \phi_Ak_4C_{MG} - z_5\frac{dk_4}{dT}C_{MG} - z_5k_4\theta_{MG} - \phi_Ak_6C_{GL} - z_5\frac{dk_6}{dT}C_{GL} - z_5k_6\theta_{GL} + \phi_{GL}k_6C_{GL} + z_6\frac{dk_6}{dT}C_{GL} + z_6k_6\theta_{GL}$$

$$\frac{d\phi_5}{dt} = \phi_{TG}k_1C_{TG} + z_1\frac{dk_1}{dT}C_{TG} + z_1k_1\theta_{TG} - \phi_{DG}k_1C_{TG} - z_2\frac{dk_1}{dT}C_{TG} - z_2k_1\theta_{TG} + \phi_{DG}k_3C_{DG} + z_2\frac{dk_3}{dT}C_{DG} + z_2k_3\theta_{DG} - \phi_{MG}k_3C_{DG} - z_3\frac{dk_3}{dT}C_{DG} - z_3k_3\theta_{DG} + \phi_{MG}k_5C_{MG} + z_3\frac{dk_5}{dT}C_{MG} + z_3k_5\theta_{MG} - \phi_Ek_1C_{TG} - z_4\frac{dk_1}{dT}C_{TG} - z_4k_1\theta_{TG}0 - \phi_Ek_3C_{DG} - z_4\frac{dk_3}{dT}C_{DG} - z_4k_3\theta_{DG} - \phi_Ek_5C_{MG} - z_4\frac{dk_5}{dT}C_{MG}0 - z_4k_5\theta_{MG} + \phi_Ak_1C_{TG} + z_5\frac{dk_1}{dT}C_{TG} + z_5k_1\theta_{TG} + \phi_Ak_3C_{DG} + z_5\frac{dk_3}{dT}C_{DG} + z_5k_3\theta_{DG} + \phi_Ak_5C_{MG} + z_5\frac{dk_5}{dT}C_{MG} + z_5k_5\theta_{MG} - \phi_{GL}k_5C_{MG} - z_6\frac{dk_5}{dT}C_{MG} - z_6k_5\theta_{MG}$$

$$\frac{d\phi_6}{dt} = -\phi_{MG}k_6C_E - z_3\frac{dk_6}{dT}C_E - z_3k_6\theta_E - \phi_Ek_6C_E - z_4\frac{dk_6}{dT}C_E - z_4k_6\theta_E - \phi_Ak_6C_E - z_5\frac{dk_6}{dT}C_E - z_5k_6\theta_E + \phi_{GL}k_6C_E + z_6\frac{dk_6}{dT}C_E + z_6k_6\theta_E$$

…….(19)

c) **Psi Equations**

$$\frac{d\Psi_1}{dt} = \Psi_{TG}k_1C_A + 2\omega_1\frac{dk_1}{dT}C_A + \omega_1k_1\theta_A - \Psi_{DG}k_1C_A - 2\omega_2\frac{dk_1}{dT}C_A - \omega_2k_1\theta_A - \Psi_Ek_1C_A - 2\omega_4\frac{dk_1}{dT}C_A - \omega_4k_1\theta_{A+}\Psi_Ak_1C_A + 2\omega_5\frac{dk_1}{dT}C_A + \omega_5k_1\theta_A$$

$$\frac{d\Psi_2}{dt} = -\Psi_{TG}k_2C_E - 2\omega_1\frac{dk_2}{dT}C_E - \omega_1k_1\theta_E + \Psi_{DG}k_2C_E + 2\omega_2\frac{dk_2}{dT}C_E + \omega_2k_2\theta_E + \Psi_{DG}k_3C_A + 2\omega_2\frac{dk_3}{dT}C_A + \omega_2k_3\theta_A - \Psi_{MG}k_3C_A - 2\omega_3\frac{dk_3}{dT}C_A - \omega_3k_3\theta_A + \Psi_Ek_2C_E + 2\omega_4\frac{dk_2}{dT}C_E + \omega_4k_2\theta_E + \Psi_Ek_3C_A + 2\omega_4\frac{dk_3}{dT}C_A + \omega_4k_3\theta_A$$

$$\frac{d\Psi_3}{dt} = -\Psi_{DG}k_4C_E - 2\omega_2\frac{dk_4}{dT}C_E - \omega_2k_4\theta_E + \Psi_{MG}k_4C_E + 2\omega_3\frac{dk_4}{dT}C_E + \omega_3k_4\theta_E + \Psi_{MG}k_5C_A + 2\omega_3\frac{dk_5}{dT}C_A + \omega_3k_5\theta_A + \Psi_Ek_4C_E + 2\omega_4\frac{dk_4}{dT}C_A + \omega_3k_3\theta_E - \Psi_Ek_5C_A - 2\omega_4\frac{dk_5}{dT}C_A - \omega_4k_5\theta_A - \Psi_{GL}k_5C_A - 2\omega_6\frac{dk_5}{dT}C_A - \omega_6k_5\theta_A$$

$$\frac{d\Psi_4}{dt} = -\Psi_{TG}k_4C_{DG} - 2\omega_1\frac{dk_2}{dT}C_{DG} - \omega_1k_2\theta_{DG} + \Psi_{DG}k_2C_{DG} + 2\omega_2\frac{dk_2}{dT}C_{DG} + \omega_2k_2\theta_{DG} - \Psi_{DG}k_4C_{TG} - 2\omega_2\frac{dk_4}{dT}C_{TG} - \omega_2k_4\theta_{TG} + \Psi_{MG}k_4C_{TG} + 2\omega_3\frac{dk_4}{dT}C_{TG} + \omega_3k_4\theta_{TG} - \Psi_{MG}k_6C_{GL} - 2\omega_3\frac{dk_6}{dT}C_{GL} - \omega_3k_6\theta_{GL} + \Psi_Ek_2C_{DG} + 2\omega_4\frac{dk_2}{dT}C_{DG} + \omega_4k_2\theta_{DG} + \Psi_Ek_4C_{TG} + 2\omega_4\frac{dk_4}{dT}C_{TG} + \omega_4k_4\theta_{TG} + \Psi_Ek_6C_{GL} + 2\omega_4\frac{dk_6}{dT}C_{GL} + \omega_4k_6\theta_{GL} - \Psi_Ak_2C_{DG} - 2\omega_5\frac{dk_2}{dT}C_{DG} - \omega_4k_4\theta_{DG} - \Psi_Ak_4C_{TG} - 2\omega_5\frac{dk_4}{dT}C_{TG} - \omega_5k_4\theta_{TG} - \Psi_Ak_6C_{GL} - 2\omega_5\frac{dk_6}{dT}C_{GL} - \omega_5k_6\theta_{GL} + \Psi_{GL}k_6C_{GL} + 2\omega_6\frac{dk_6}{dT}C_{GL} + \omega_6k_6\theta_{GL}$$

$$\frac{d\Psi_5}{dt} = \Psi_{TG}k_1C_{MG} + 2\omega_1\frac{dk_1}{dT}C_{MG} + \omega_1k_1\theta_{MG} - \Psi_{DG}k_1C_{MG} - 2\omega_2\frac{dk_1}{dT}C_{MG} - \omega_2k_1\theta_{MG}$$
$$+ \Psi_{DG}k_3C_{DG} + 2\omega_2\frac{dk_3}{dT}C_{DG} + \omega_2k_3\theta_{DG} - \Psi_{MG}k_3C_{DG} - 2\omega_3\frac{dk_3}{dT}C_{DG}$$
$$- \omega_3k_3\theta_{DG} + \Psi_{MG}k_5C_{TG} + 2\omega_3\frac{dk_5}{dT}C_{TG} + \omega_3k_5\theta_{TG} - \Psi_Ek_1C_{MG}$$
$$- 2\omega_4\frac{dk_1}{dT}C_{MG} - \omega_4k_1\theta_{MG} - \Psi_Ek_3C_{DG} - 2\omega_4\frac{dk_3}{dT}C_{DG} - \omega_4k_3\theta_{DG}$$
$$- \Psi_Ek_5C_{TG} - 2\omega_4\frac{dk_5}{dT}C_{TG} - \omega_4k_5\theta_{TG} + \Psi_Ak_1C_{MG} + 2\omega_5\frac{dk_1}{dT}C_{MG}$$
$$+ \omega_5k_1\theta_{MG} + \Psi_Ak_3C_{DG} + 2\omega_5\frac{dk_3}{dT}C_{DG} + \omega_5k_3\theta_{DG} + \Psi_Ak_5C_{TG}$$
$$+ 2\omega_5\frac{dk_5}{dT}C_{TG} + \omega_5k_5\theta_{TG} - \Psi_{GL}k_5C_{TG} - 2\omega_6\frac{dk_6}{dT}C_{TG} - \omega_6k_5\theta_{TG}$$

$$\frac{d\Psi_6}{dt} = -\Psi_{MG}k_6C_E - 2\omega_3\frac{dk_6}{dT}C_E - \omega_3k_6\theta_E + \Psi_Ek_6C_E + 2\omega_4\frac{dk_6}{dT}C_E + \omega_4k_6\theta_E$$
$$- \Psi_Ak_6C_E - 2\omega_5\frac{dk_6}{dT}C_E - \omega_5k_6\theta_E + \Psi_{GL}k_6C_E + 2\omega_6\frac{dk_6}{dT}C_E + \omega_6k_6\theta_E$$

……(20)

The theta (18), phi (19) and psi (20) equations are now substituted in the Hamiltonian derivative equation, which gives the dH/dT values. Finally, a numerical method such as RKF is used to integrate the system of equations represented with initial conditions: $h_i(t_0)$ = [0;0;0;0;0;0]. Similarly, to compute i, backward integration and using again RKF method is also used to solve the system of equation, but in this case, the boundary conditions are expressed as: [0; 0; 0; 0; 0; 0].

Once the Hamiltonian Plot(dH/dT) plot has been obtained, we proceed with plotting the optimal temperature profile (T_r versus time).

CHAPTER 4: SIMULATION

MATLAB SIMULATIONS

From the start of the project we, have formed the ordinary differential equations required to form the optimum temperature profile using MATLAB. The final optimum temperature profile has also been formed using the odes. We have solved all the odes using MATLAB codes which are as follows:

1) Plot of Concentration vs Time (dC/dt) :

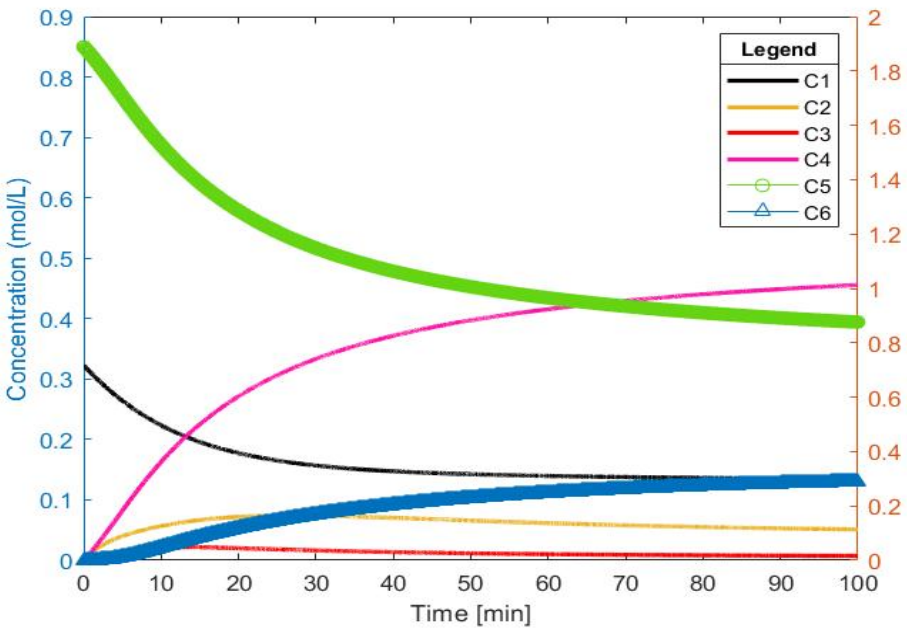

Fig 4.1. Concentration profile at constant temperature for 100 mins (dC/dt)

The objective of Fig. 4.1 is to illustrate that as the reaction progresses, there is a steady rise in the concentration of the desired product, methyl ester (biodiesel). Beyond the 65-minute mark, there is no substantial alteration in these concentrations. This trend is also observed in the other components, where their values remain relatively constant after reaching this specific point in time.

Likewise, by employing the RKF method and conducting backward integration in MATLAB, the equations for dz/dt (Fig. 4.2) and dω/dt (Fig. 4.3) are solved. Subsequently, the plots of z versus time and ω versus time is generated using the data obtained from the previous plot of dC/dt, which is derived from the concentration(C) data.

2) Plot of z_i vs time (dz/dt):

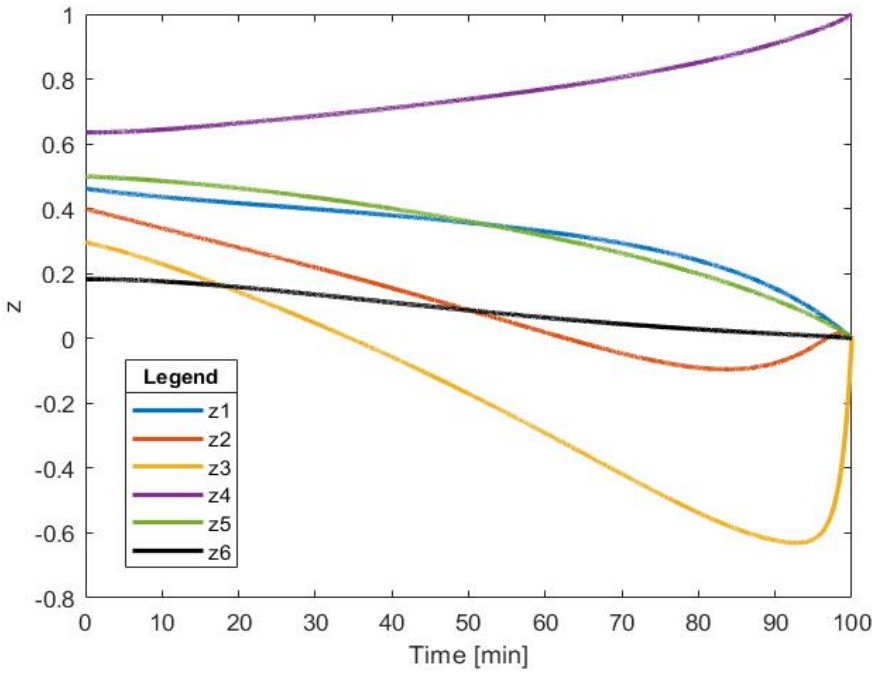

Fig. 4.2 Adjoint variable(z_i) vs. Time Plot (for 100 mins)

3) Plot of ω_i vs time (dω/dt):

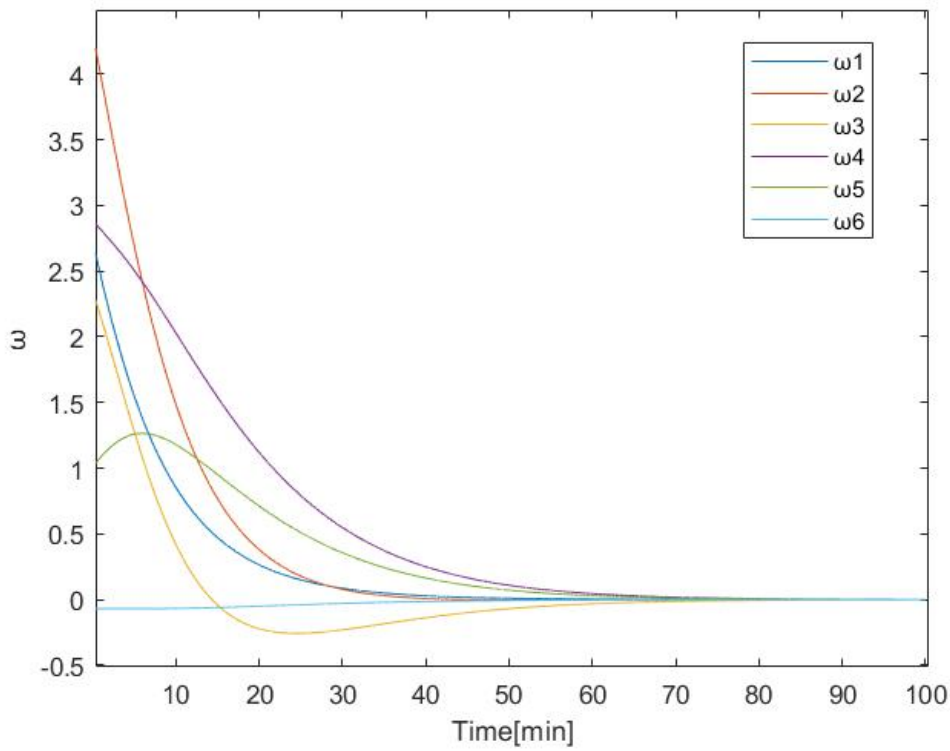

Fig. 4.3 Adjoint variable(ω_i) vs. Time Plot (for 100 mins)

From the theory, we understand that theta phi and psi are required to find dH/dt. This is why dθ/dt (Fig. 4.4), dΦ/dt (Fig. 4.5) and dΨ/dt (Fig. 4.6) are solved using RKF method again using forward and backward integration to find theta phi and psi.

4) **<u>Theta vs Time plot (dθ/dt):</u>**

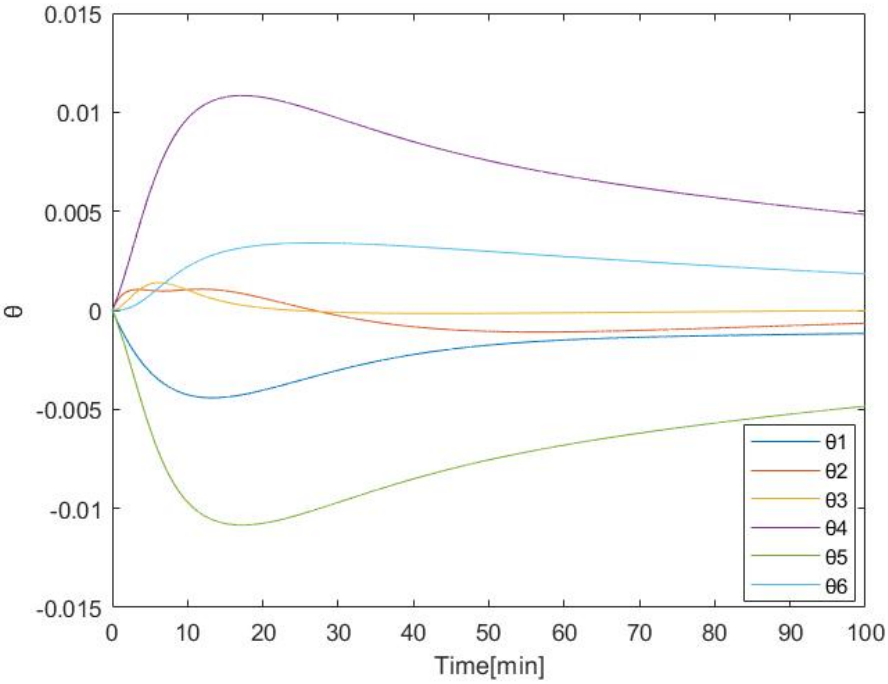

Fig. 4.4 Theta(θ) vs Time plot for 100 min

This plot, Fig 4.4 is used as an input values to find phi. This is why dΦ/dt (Fig. 4.5) is solved using RKF method again with forward integration to find theta which in turn helps to find phi and psi.

5) Phi vs Time plot (dΦ/dt):

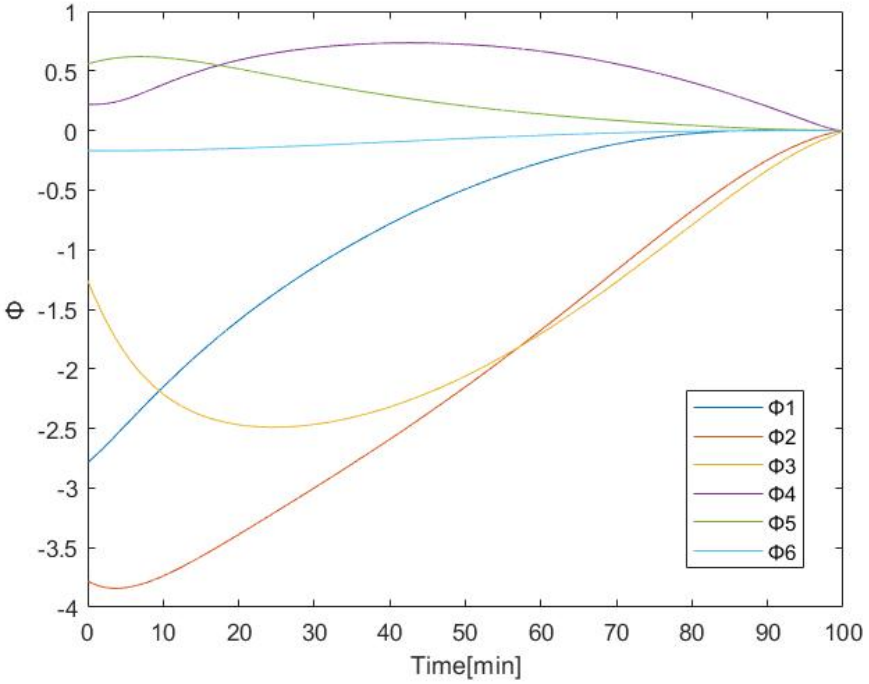

Fig. 4.5 Phi(Φ) vs Time plot for 100 min

6) Psi vs Time plot (dΨ/dt):

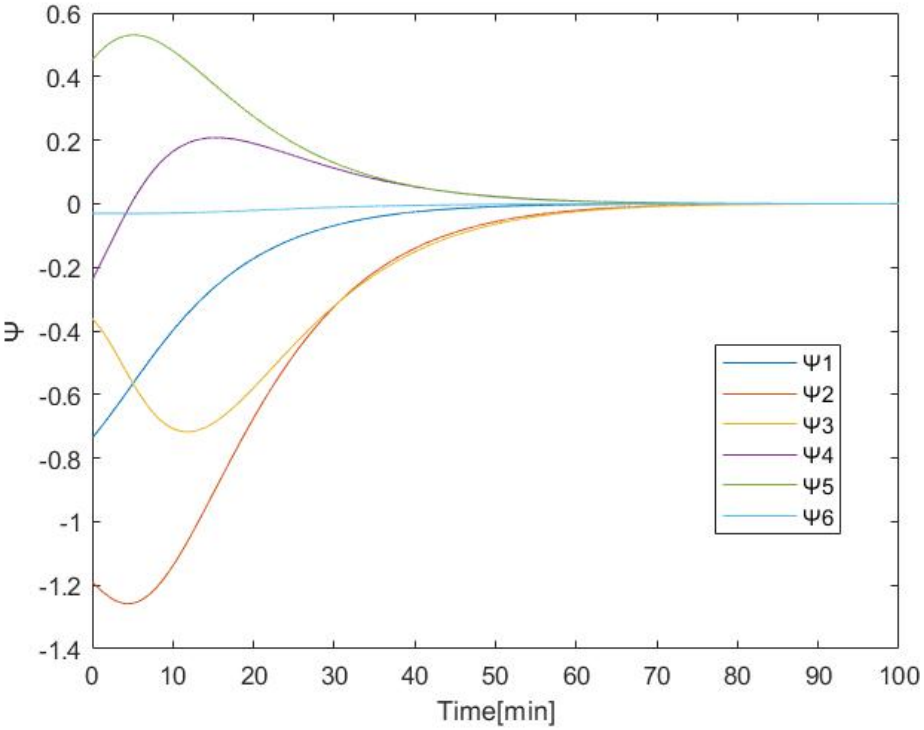

Fig. 4.6 Psi(Ψ) vs Time plot for 100 mins

7) Hamiltonian Plot(dHdT/dt):

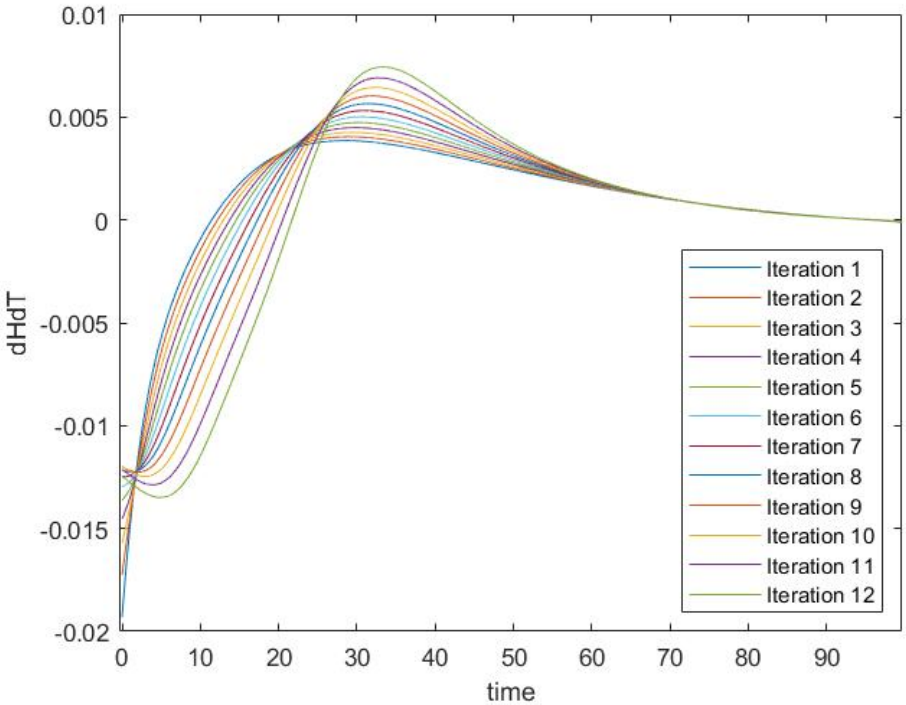

Fig. 4.7 Iterative dH/dT vs Time plot for 100 min

8) Reaction temperature vs Time plot (T_r/dt):

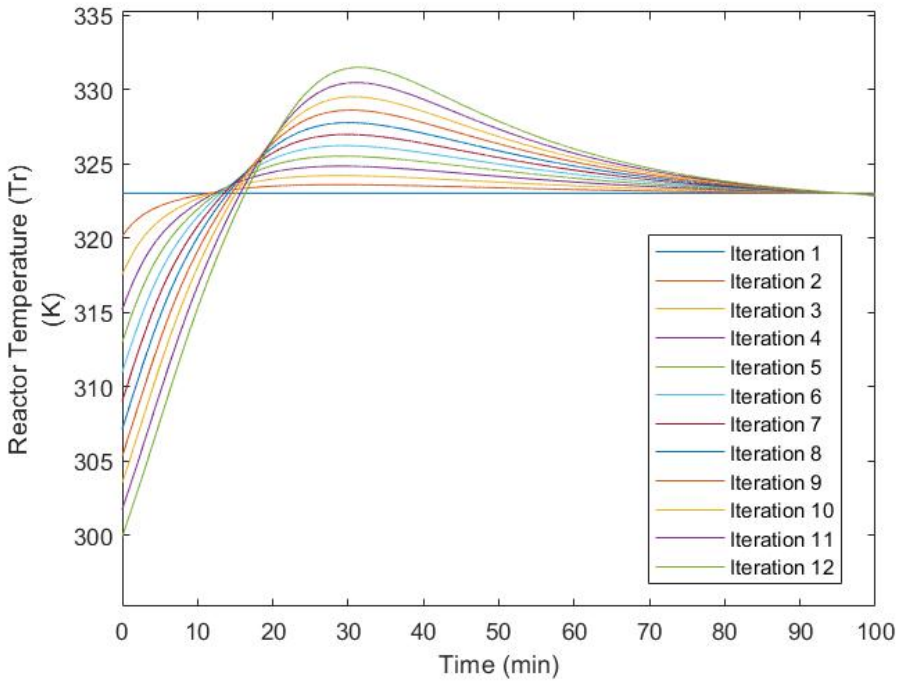

Fig. 4.8 Iterative Reaction temperature (T_r) vs Time plot for 100 min

Numerical results showed that 12 iterations were enough to reach the max concentration of methyl ester in Fig 4.7. The optimal temperature profile was obtained using the iterative process and is depicted in Fig. 4.8.

We will consider the iteration where the tolerance has been met i.e., the final iteration (12th) as the optimal profile for biodiesel production in a batch reactor under stochastic control.

9) Optimal Profile for maximum biodiesel production (Stochastic Control):

Fig. 4.9 depicts the final optimal temperature profile of the batch reactor for biodiesel production considering the inherent uncertainties by using the stochastic control approach. Fig. 4.8 shows the formation of optimal profile from 323 K to final profile formation.

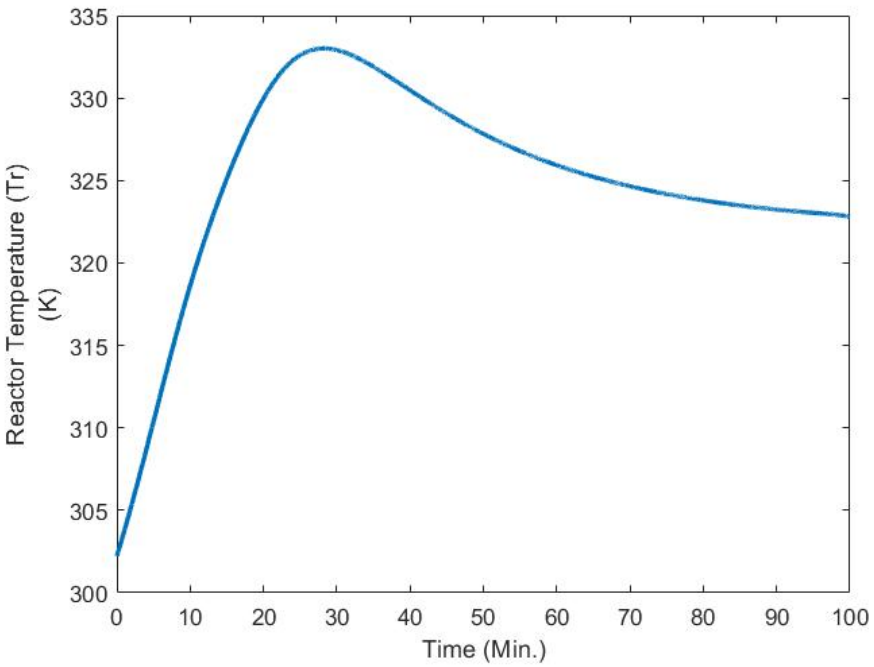

Fig. 4.9 Optimal temperature profile for maximum biodiesel production (Stochastic Control)

CHAPTER 5: RESULTS

1. DETERMINISTIC VS STOCHASTIC CASES

To analyse the difference between the stochastic control case and deterministic case, we shall run simulations without the uncertainty and ω variable. Since the ω variable accounts for the uncertainties in the biodiesel production, it does not exist in the deterministic control approach.

Optimal Profile for maximum biodiesel production (Deterministic Control):

Fig 5.1 depicts optimal temperature profiles for biodiesel production using the deterministic method.

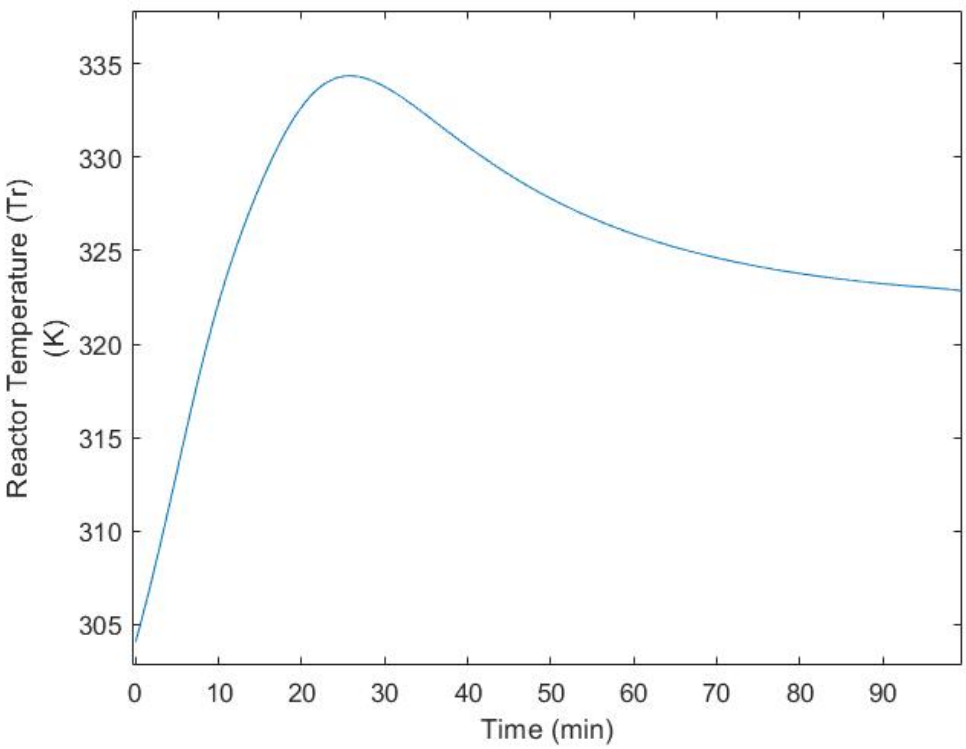

Fig. 5.1 Optimal temperature profile for maximum biodiesel production (Deterministic Control)

Comparison between deterministic and stochastic temperature profiles:

Fig 5.2 depicts the difference between the optimal temperature profiles of stochastic and deterministic cases for biodiesel production.

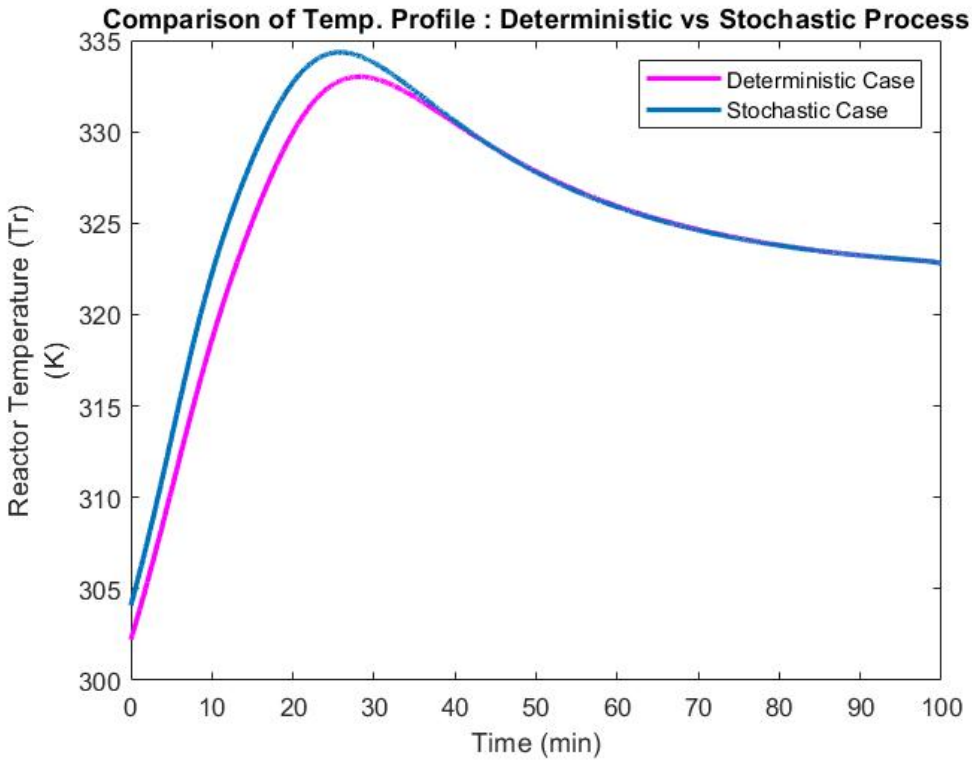

Fig. 5.2 Comparison of Deterministic vs Stochastic Control

2. ANALYSIS OF THE RESULTS

Improved Yield and Quality: One of the key findings of the project is that the stochastic control strategy significantly improves the yield and quality of the biodiesel product compared to open-loop control. This improvement is attributed to the ability of stochastic control to adapt to the variability in feed composition and reactor operation. The closed-loop approach ensures that the process operates optimally, leading to a higher concentration of methyl ester (biodiesel) in the final product.

Sensitivity to Uncertainty: This study demonstrates that the stochastic control strategy is sensitive to the uncertainties associated with feed composition. By using stochastic differential

equations (SDEs) and Ito's Lemma, the model accounts for the variability in feedstock composition.

Comparison with Deterministic Control: The study compares the stochastic control approach with deterministic control. The results show that the stochastic control strategy consistently outperforms deterministic control in terms of both yield and cost-efficiency. This highlights the advantage of considering uncertainty explicitly in the control strategy.

Optimal Temperature Profile: The study reveals that the optimal temperature profile for the transesterification reactor is a critical factor in maximizing the expected concentration of methyl ester. The temperature profile is obtained iteratively by solving the Hamiltonian Equation, and it adapts to the changing conditions in the reactor. This dynamic optimization contributes significantly to process efficiency.

Real-World Applicability: While the study is based on mathematical modelling and simulations, it has practical implications for the biodiesel production industry. It demonstrates the potential for implementing closed-loop control systems that can adjust production parameters in real-time to respond to uncertainties, ultimately improving product quality and yield.

In summary, the research paper underscores the effectiveness of stochastic control strategies in addressing uncertainties in biodiesel production. It provides valuable insights into the potential for optimizing batch reactor operations under variable conditions, with a focus on maximizing biodiesel production efficiency and quality.

CHAPTER 6: CONCLUSION

In conclusion, this project focused on the optimal control of biodiesel production in a batch reactor under uncertainty in the feed composition. The study proposed a stochastic control approach using the stochastic maximum principle and Ito's Lemma. The stochastic control approach helped to account for the variability in the feed composition and optimize the production process while minimizing the cost.

The results obtained from the study demonstrated the effectiveness of the stochastic control approach in optimizing the biodiesel production process. The study showed that by incorporating the uncertainty in the feed composition, the expected concentration of methyl ester in the biodiesel product can be estimated more accurately, leading to improved control over the production process. Stochastic control strategies significantly increased biodiesel yield and improved product quality compared to the deterministic case. The system's adaptability to uncertainties in feedstock composition was highlighted.

In summary, this project provides insights into the optimal control of biodiesel production under uncertainty, highlighting the importance of stochastic control approaches and optimization techniques in improving the efficiency of chemical processes.

REFERENCES

[1] Urmila Diwekar, Pahola T. Benavides, 'Optimal control of biodiesel production in a batch reactor Part II: Stochastic control', 2011

[2] Urmila Diwekar, Pahola T. Benavides, 'Optimal control of biodiesel production in a batch reactor Part I: Deterministic control', 2011

[3] Prajwal Shettigar J, 'Data-Driven Modelling of a Pilot Plant Batch Reactor and Validation of a Nonlinear Model Predictive Controller for Dynamic Temperature Profile Tracking', 2021

[4] Prajwal Shettigar J, 'Wiener-Neural-Network-Based Modelling and Validation of Generalized Predictive Control on a Laboratory-Scale Batch Reactor', 2022

PROJECT DETAILS

Student Details			
Student 1 Name	**TANISH J**		
Register Number	190932190	Section	B
Email Address	j.tanish@gmail.com	Phone No (M)	9113079025
Project Details			
Project Title	**Optimal temperature profile formation for Biodiesel production in a batch reactor with uncertainty - Stochastic Control**		
Project Duration	9 months	Date of reporting	09-01-2023
Date of Synopsis Submission	11-04-2023	Date of Mid-term Presentation	14-06-2023
Date of Final Report Submission	05-09-2023	Date of Final Presentation	06-09-2023
Organization Details			
Organization Name	**Manipal Institute of Technology**		
Full postal address with pin code	Eshwar Nagar, Manipal-576104		
Website address	www.manipal.edu		
Location of work	Department of Instrumentation & Control Engineering		
Project Guide Details			
Project Guide Name	**Dr.I. Thirunavukkarasu**		
Designation	Professor		
Full contact address with pin code	Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal, Karnataka, India – 576 104		
Email address	it.arasu@manipal.edu	Phone No (M)	9740731983
Internal Guide Details			
Faculty Name	**Dr.I. Thirunavukkarasu**		
Full contact address with pin code	Department of Instrumentation and Control Engineering, Manipal Institute of Technology, Manipal, Karnataka, India – 576 104		
Email address	it.arasu@manipal.edu		